

Digital Image Processing:

2 Digital Imaging Fundamentals

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Contents

This lecture will cover:

1. The human visual system
2. Light and the electromagnetic spectrum
3. Image sensing and acquisition
4. Image sampling, quantization and representation
5. Image resolution

1 Human Visual System

The best vision model we have!

- ✓ Basic understanding of human visual perception is important;
- ✓ The mechanics and parameters related to how images are formed in the eye;
- ✓ The physical limitations of human vision in terms of factors that also are used in digital image processing;
- ✓ The factors how human and electronic imaging compare in terms of resolution and ability to adapt to changes in illumination.

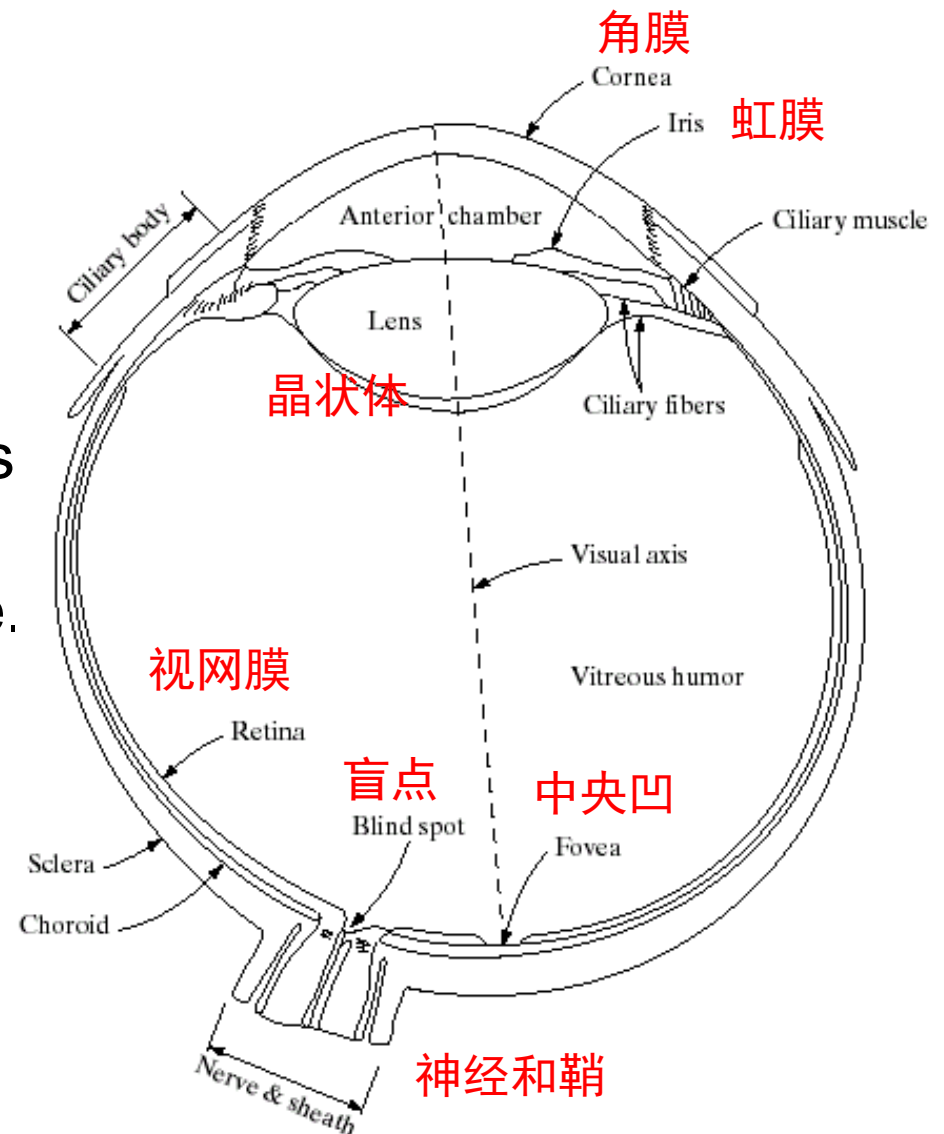
1.1 Structure Of The Human Eye

Lens: focuses light from objects onto the retina(视网膜).

- can be contracted
- zoom (to a plane)
- shape of lens is varied to focus on objects at different distances
- IR and UV light are absorbed by proteins in the lens structure.

Pupil:

- Open varies from 2-8mm
- Regulates the amount of light reaching the retina.
- Same as aperture of a camera.

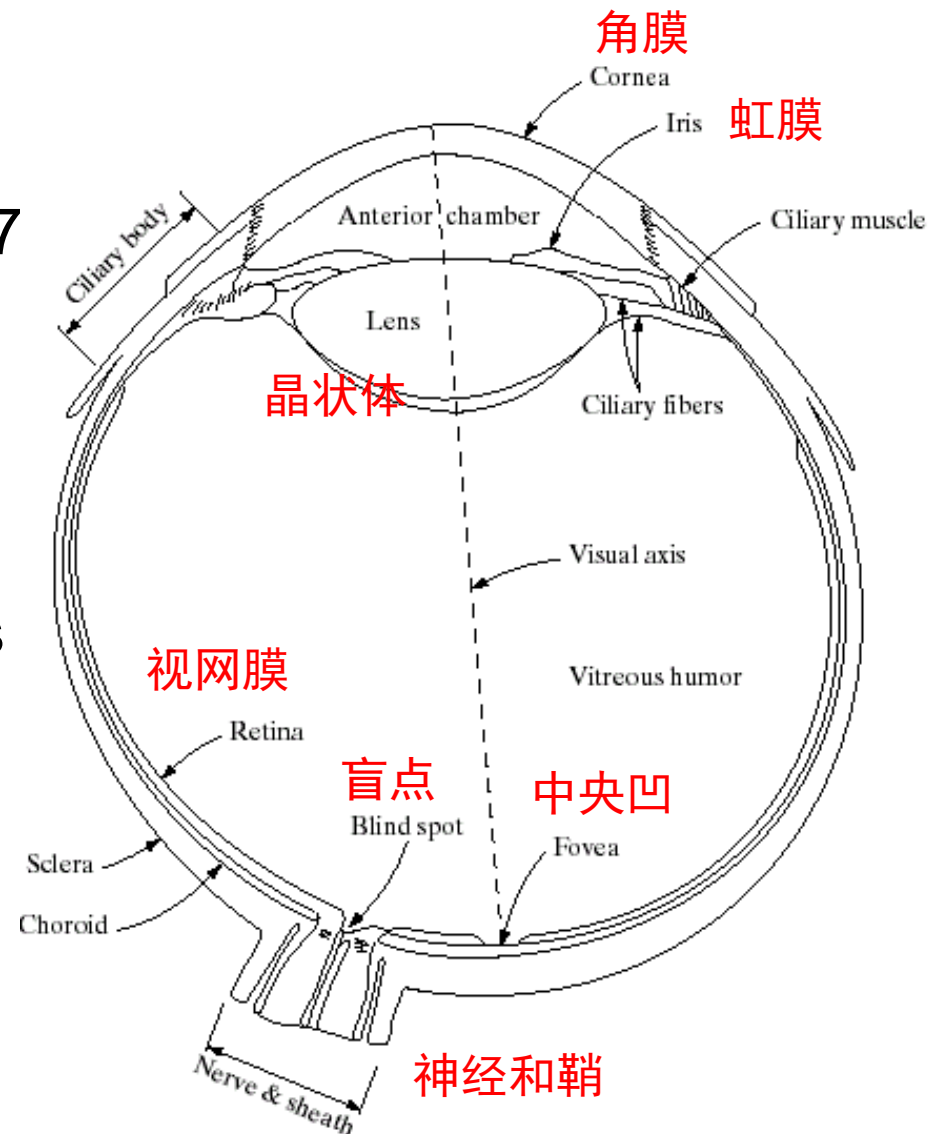


1.1 Structure Of The Human Eye

◆ Retina is covered with light receptors called *cones* 锥状体 (6-7 million) and *rods* 杆状体 (75-150 million).

□ Rods are more spread out and are sensitive to low levels of **illumination**.

□ Cones are concentrated around the fovea, very sensitive to **color**.



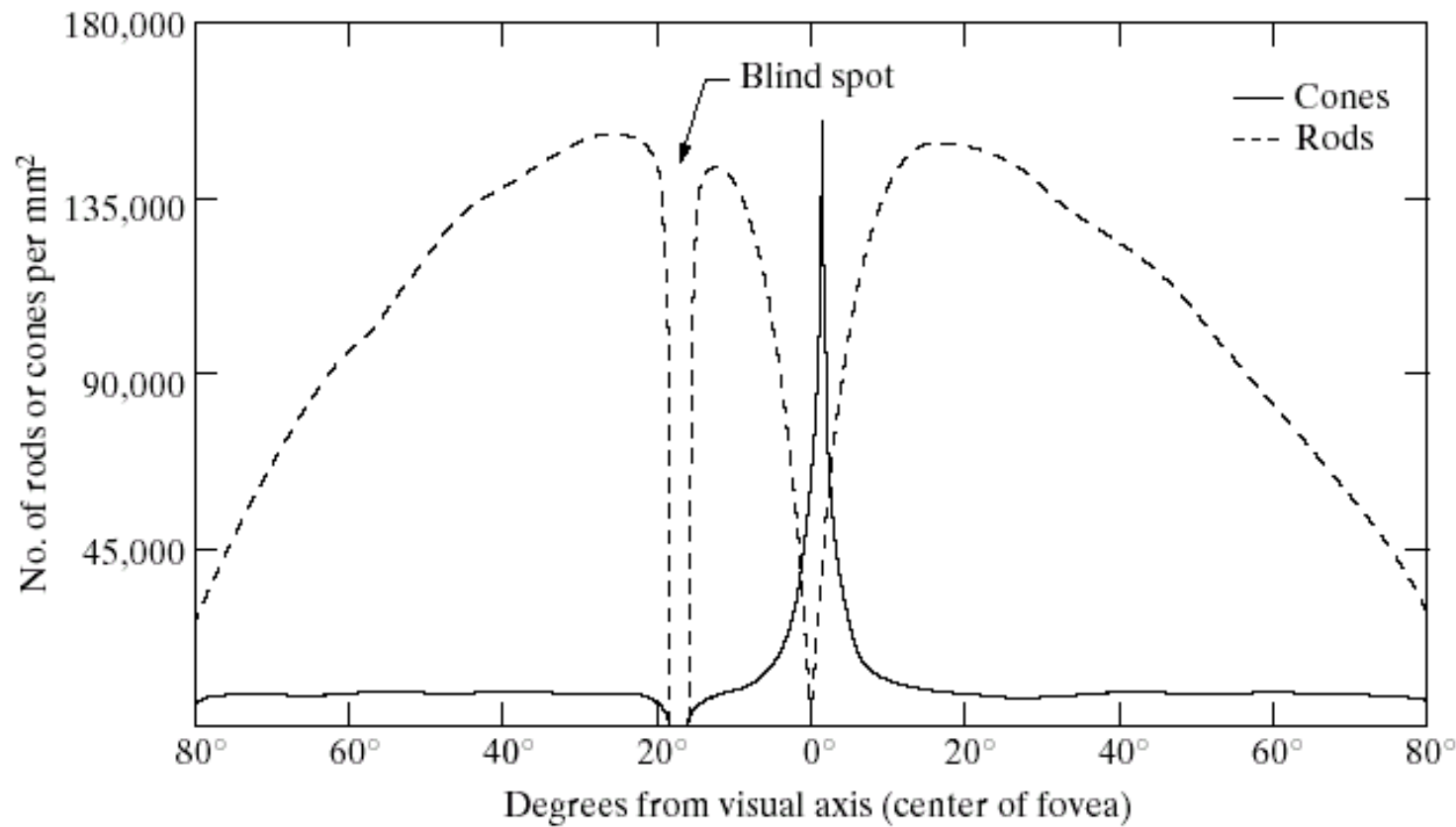


FIGURE 2.2
Distribution of
rods and cones in
the retina.

Woman or Man?

Blind-Spot Experiment

- Draw an image similar to that below on a piece of paper (the dot and cross are about 15cm apart);

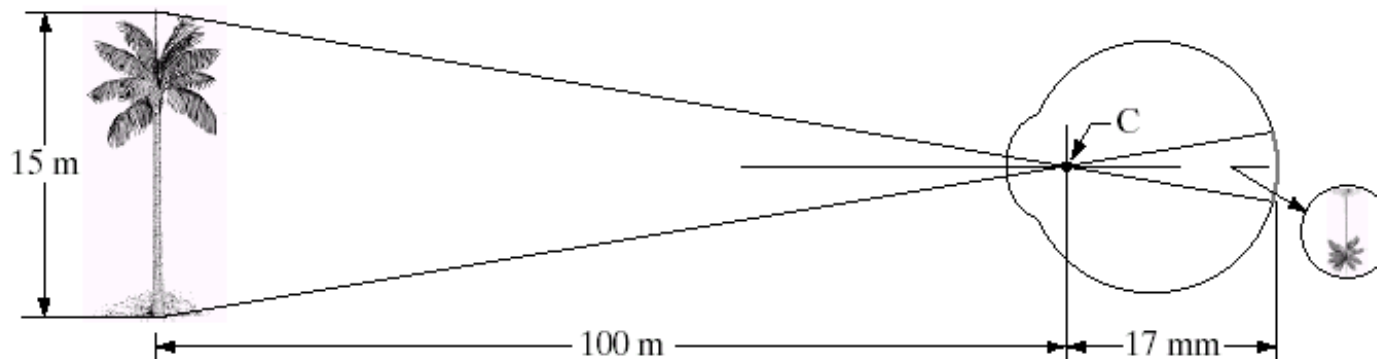


- Close your right eye and focus on the cross with your left eye. Hold the image about 50cm away from your face and move it slowly towards you.
- What's happened? The dot should disappear!

1.2 Image Formation In The Eye

Far and near: Muscles within the eye can be used to change the **shape of the lens** allowing us focus.

How: An image is focused onto the retina causing **rods and cones to become excited** which ultimately send signals to the brain.



Focal length: 14-17mm

Length of tree image $\approx 2.55\text{mm}$

For distant objects ($>3\text{m}$), lens exhibits the least refractive power (flattened)

For nearby objects ($<1\text{m}$), lens is most strongly refractive (curved)

1.3 Brightness Adaptation & Discrimination

Lightness Perception: Objective Quantities

- *Luminance* is the amount of visible light that comes to the eye from a surface.
- *Illuminance* is the amount of light incident on a surface.
- *Reflectance* (also called *albedo*) is the proportion of incident light that is reflected from a surface.
 - varies from 0% to 100% where 0% is ideal black and 100% is ideal white. In practice, typical black paint is about 5% and typical white paint about 85%.

Image Formation Model

$$f(x,y)=i(x,y)r(x,y)$$

$$0 < f(x,y) < \infty$$

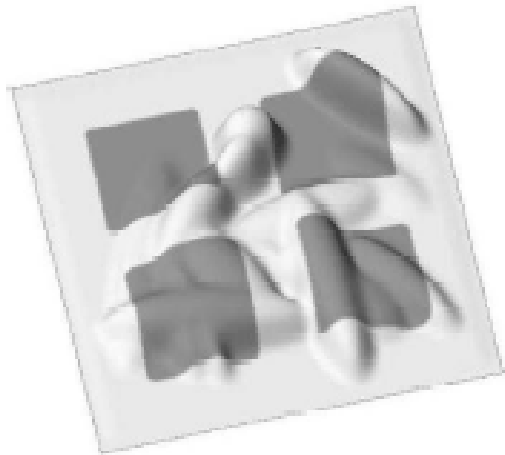
luminance – proportional to energy
radiated by a physical source

$$0 < i(x,y) < \infty$$

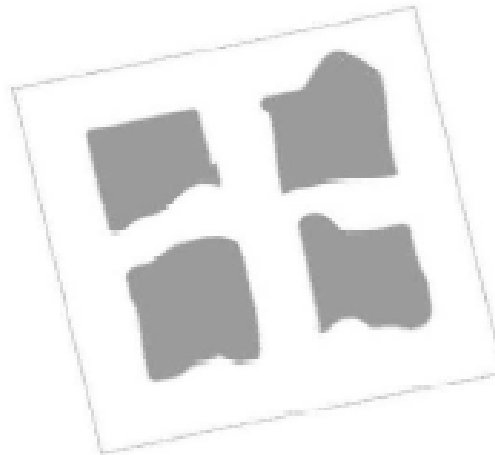
Illumination

$$0 < r(x,y) < 1$$

reflectance



$f(x,y)$



$r(x,y)$



$i(x,y)$

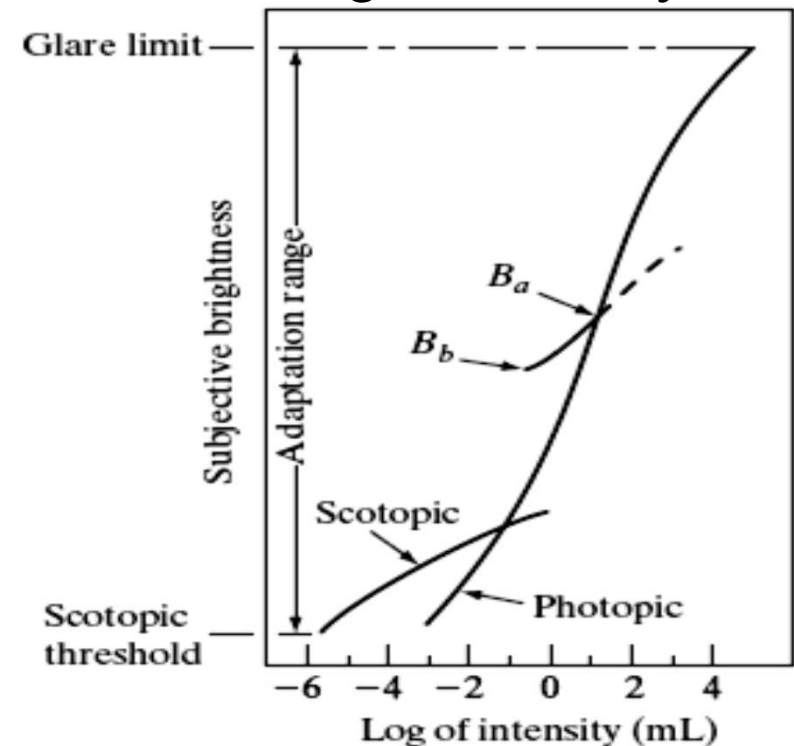
1.3 Brightness Adaptation & Discrimination

Lightness Perception: Subjective Quantities

- Lightness is the **perceived reflectance** of a surface. It represents the visual system's attempt to extract reflectance based on the luminance in the scene.
- Brightness is the perceived intensity of light coming from the image itself, rather than any property of the portrayed scene. Brightness is sometimes defined as **perceived luminance**.

1.3 Brightness Adaptation & Discrimination

- The human visual system can perceive approximately 10^{10} different light intensity levels.
- At any one time we can only discriminate between a much smaller number – *brightness adaptation*.
- Subjective brightness is a log function of the light intensity.
- **Brightness discrimination**
 - poor at low levels of illumination,
 - better with increasing illumination

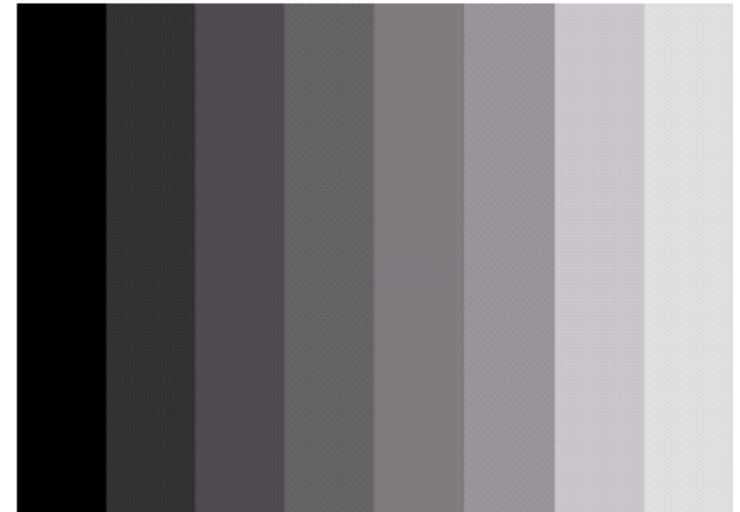


Brightness Adaptation & Discrimination (cont...)

Perceived brightness is not a simple function of intensity!

(1) Mach bands

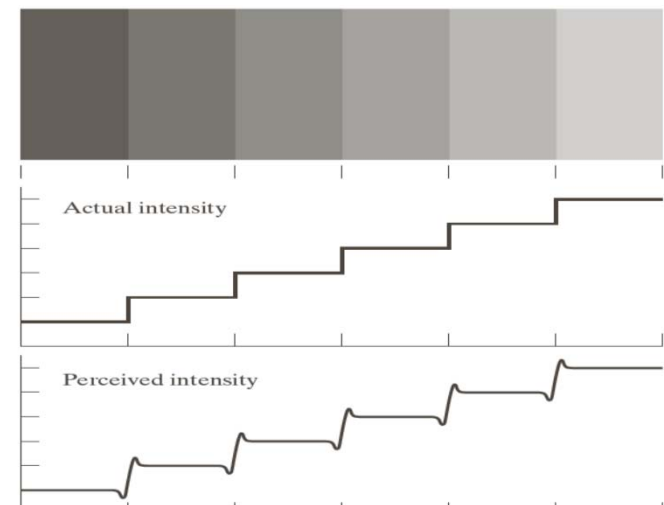
stripes appear darker near a more intense stripe (and vice versa) caused by inhibitory neural connections



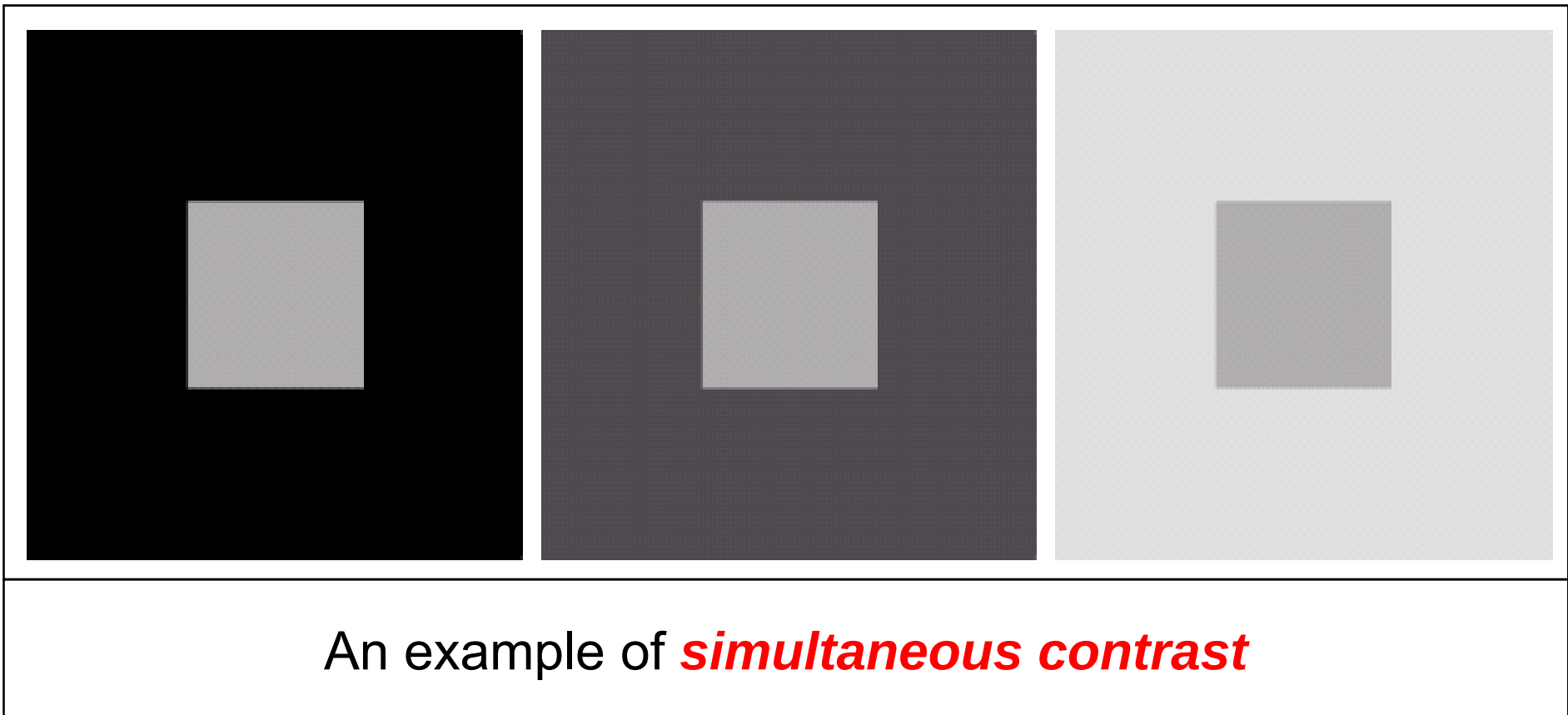
An example of Mach bands

(2) simultaneous contrast

the *perceived intensity* of a region is related to the light intensities of the **regions surrounding** it.

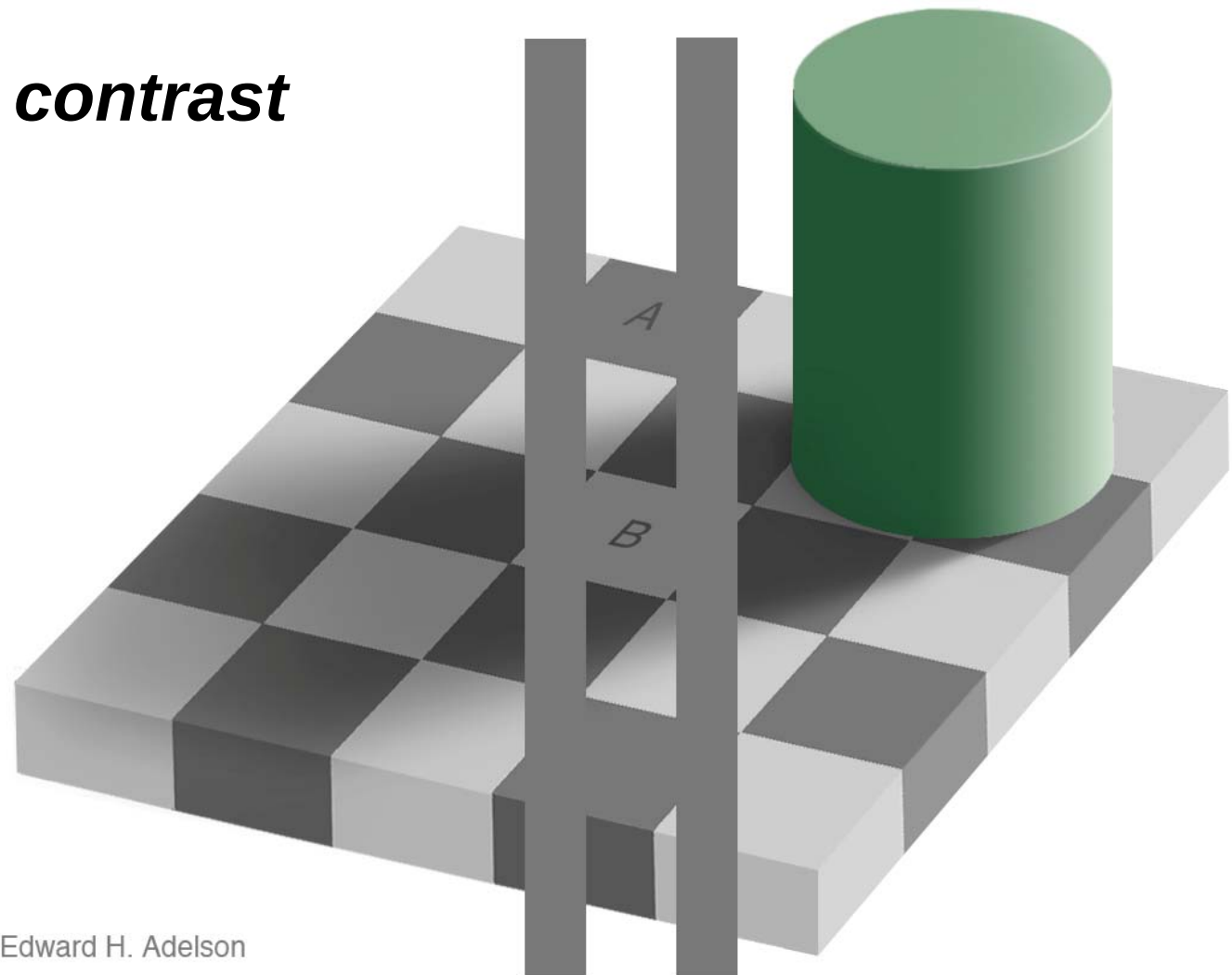


Brightness Adaptation & Discrimination (cont...)



Brightness Adaptation & Discrimination (cont...)

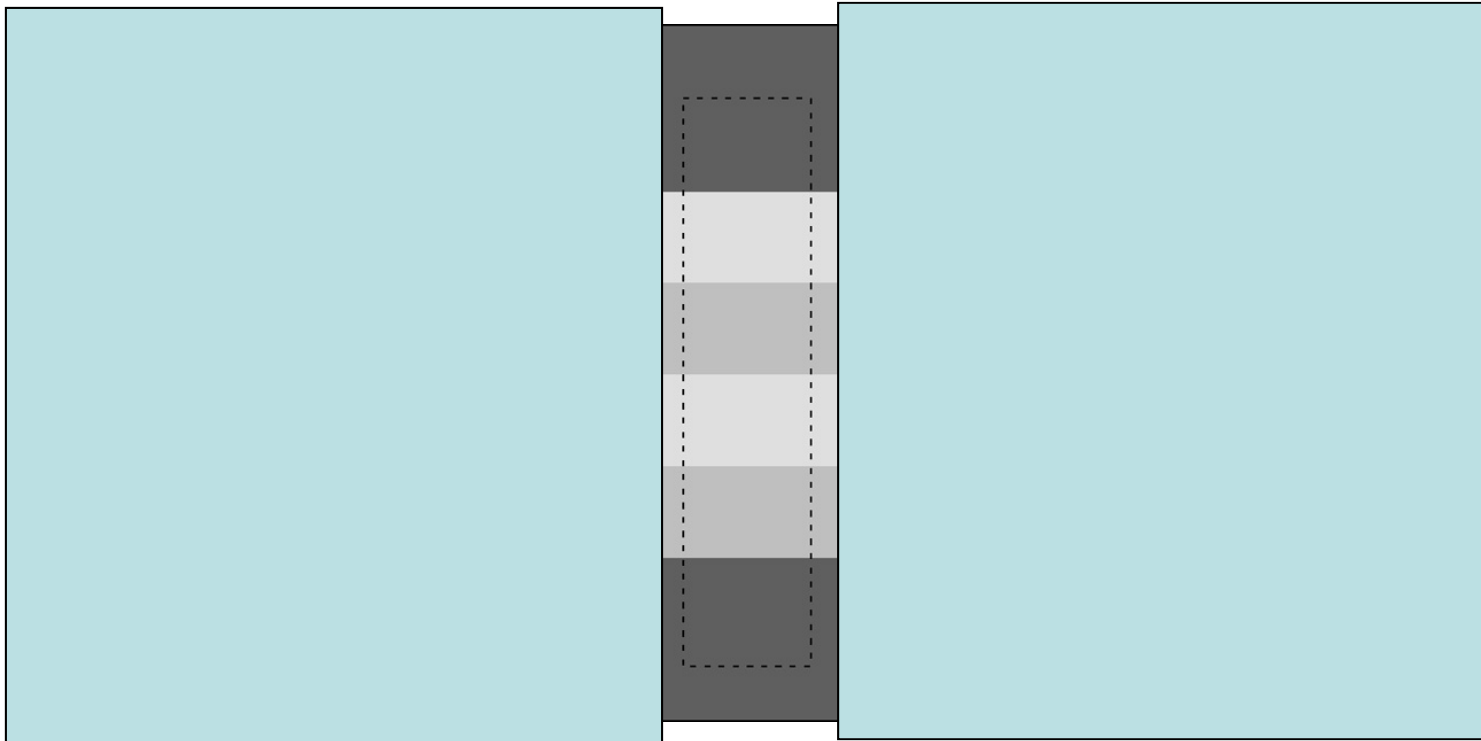
simultaneous contrast



Edward H. Adelson

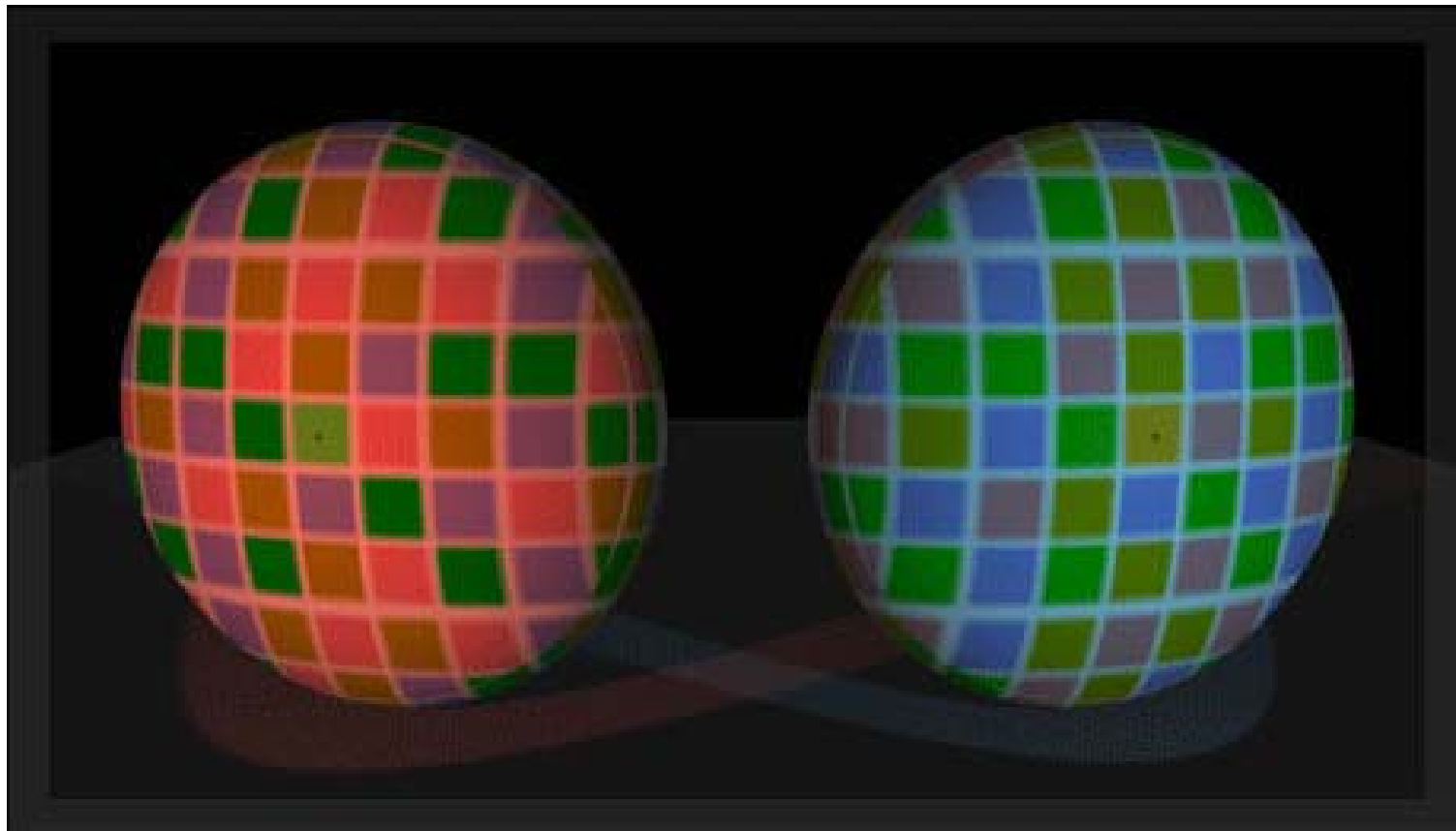
For more great illusion examples take a look at: <http://web.mit.edu/persci/gaz/>

simultaneous contrast



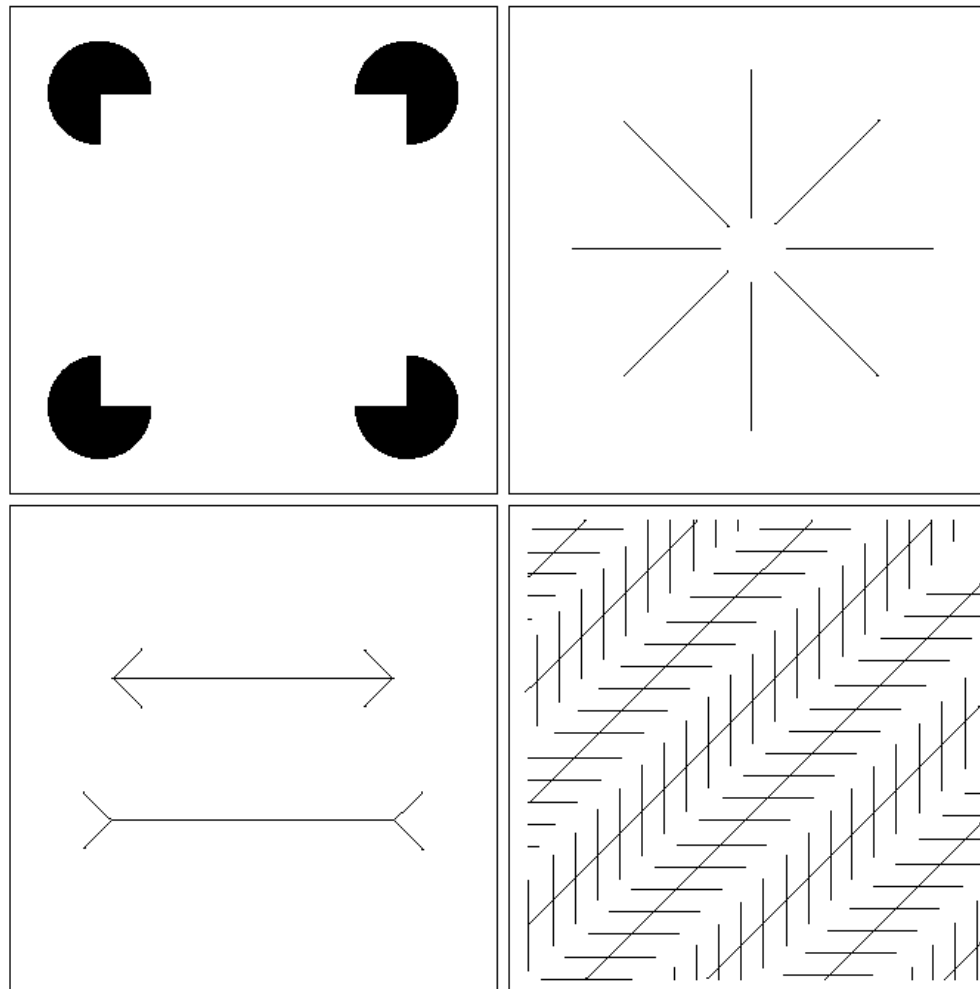
If we cover the right side of the figure and view the left side, it appears that the stripes are due to paint (**reflectance**). If we cover the left side and view the right, it appears that the stripes are due to different lighting on the stair steps (**illumination**).

simultaneous contrast



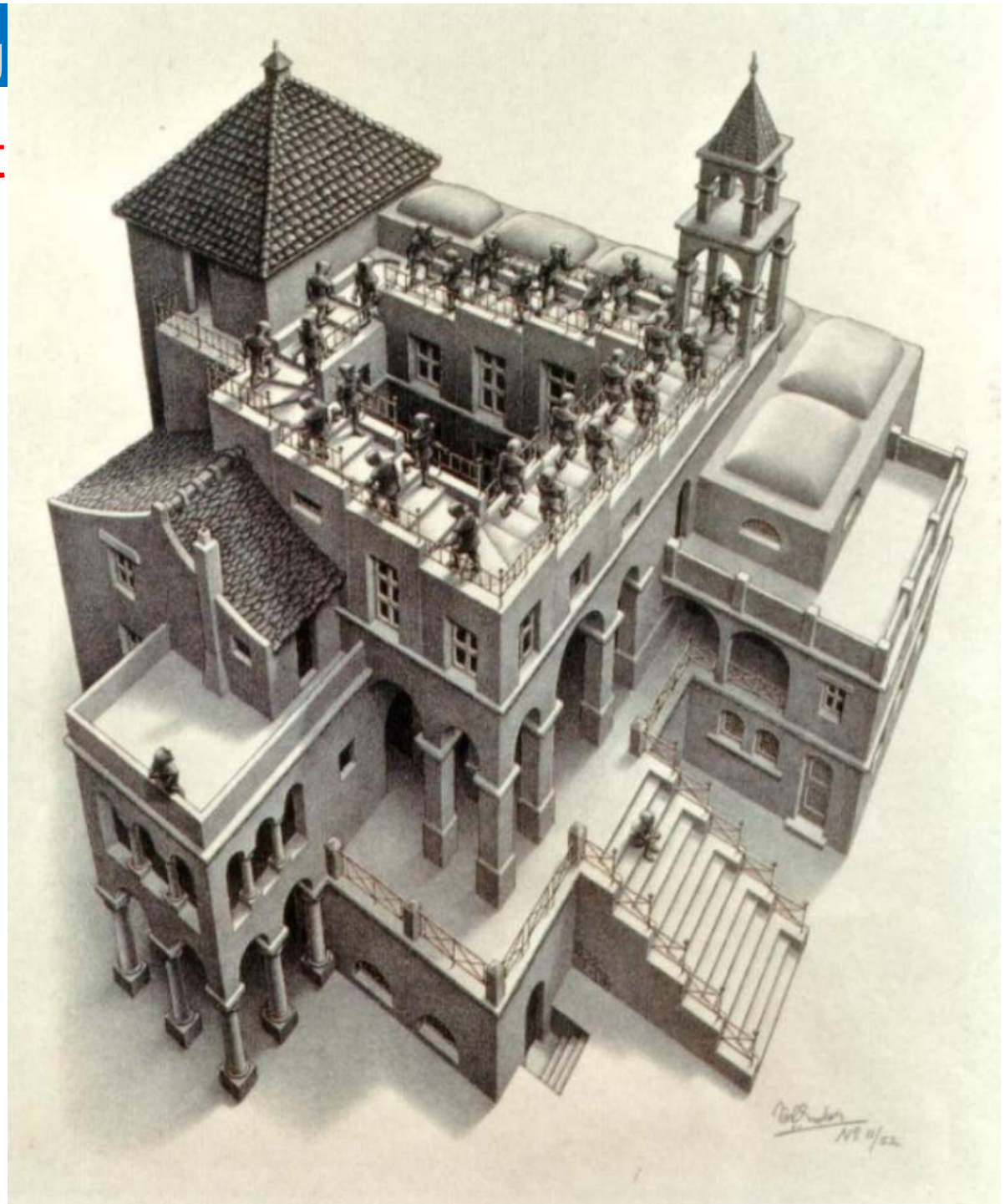
Optical Illusions

Our visual systems play lots of interesting tricks on us



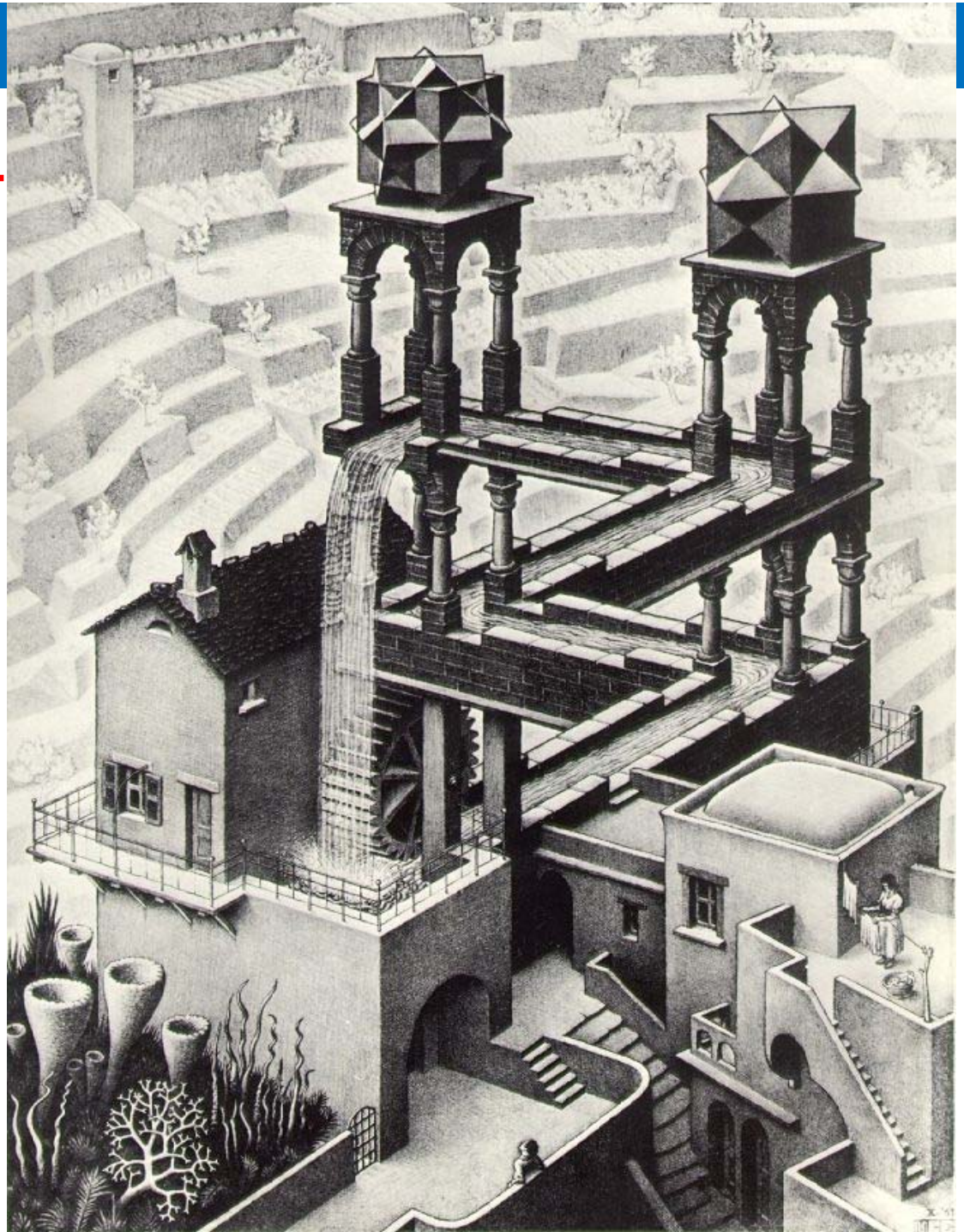
Digital Image Processing

Optical Illusions (cont)

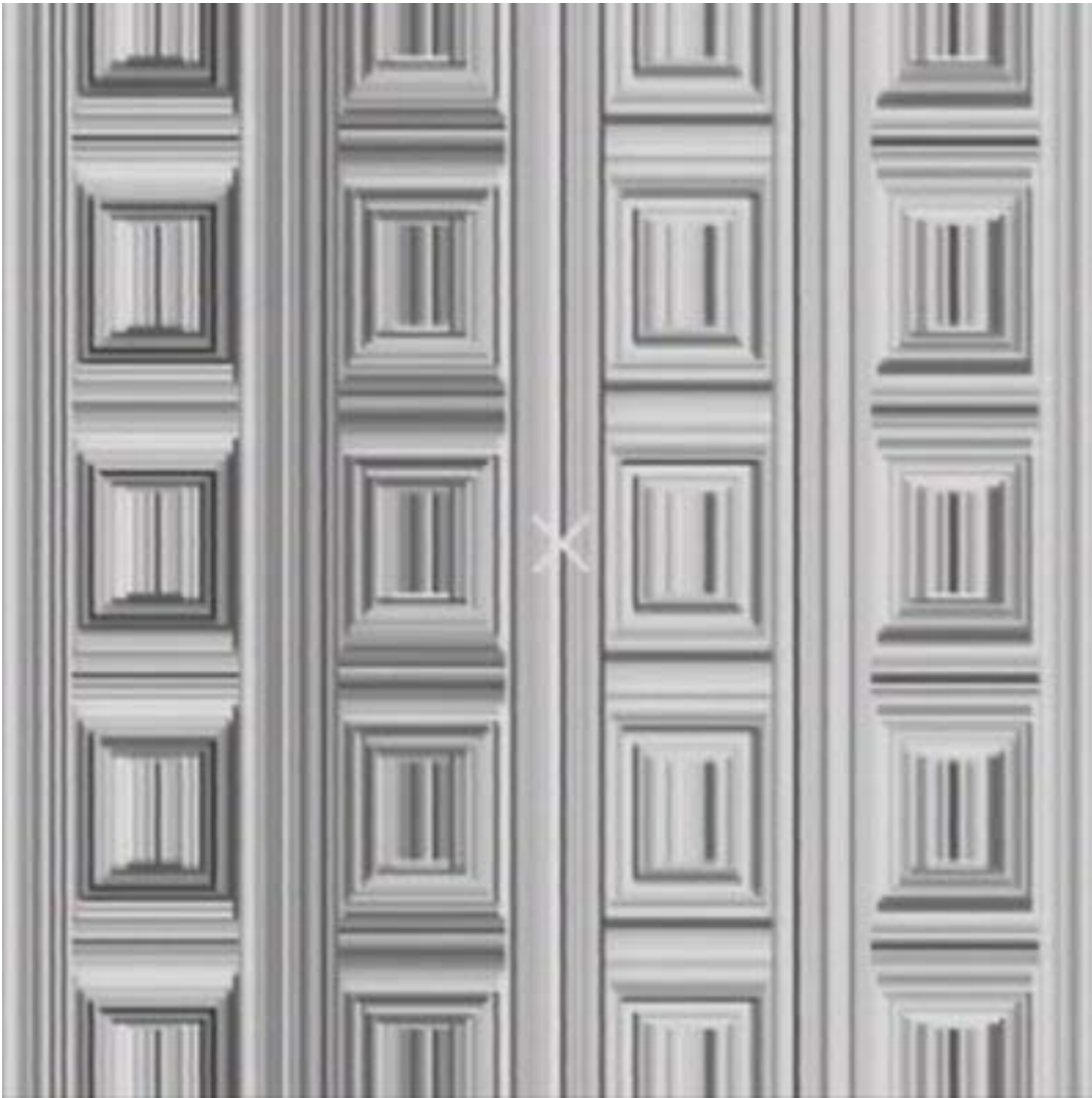


Digital Image Processing

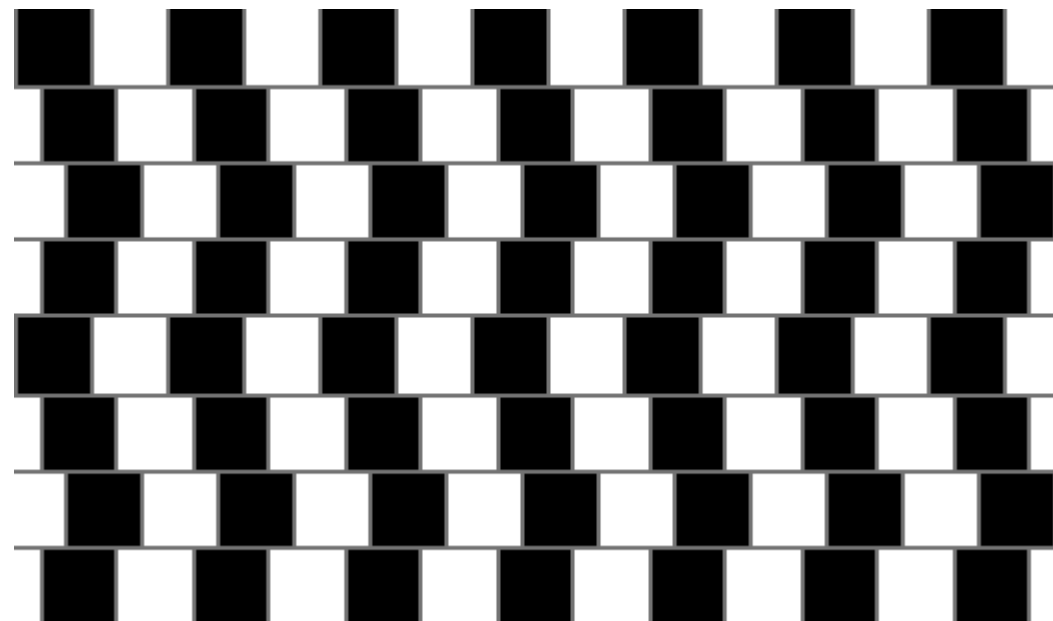
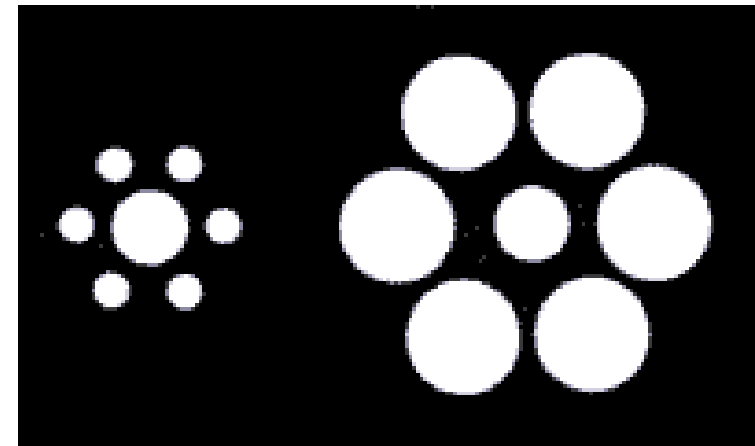
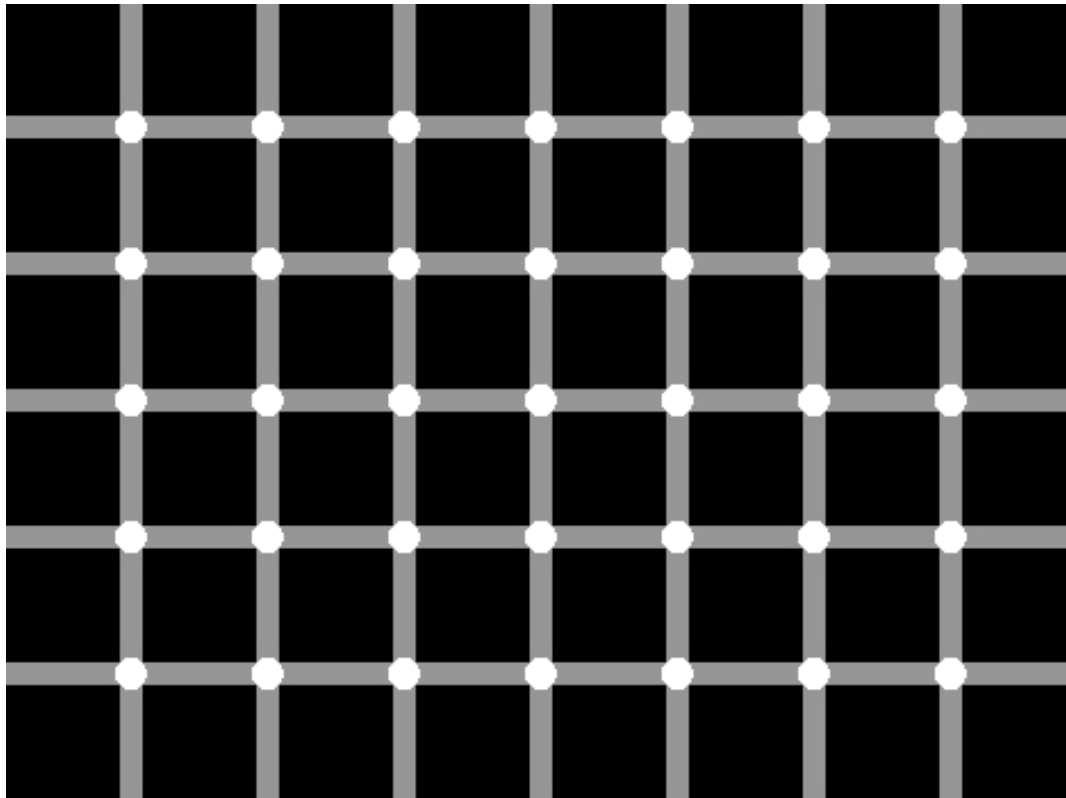
Optical Illusions (cont...

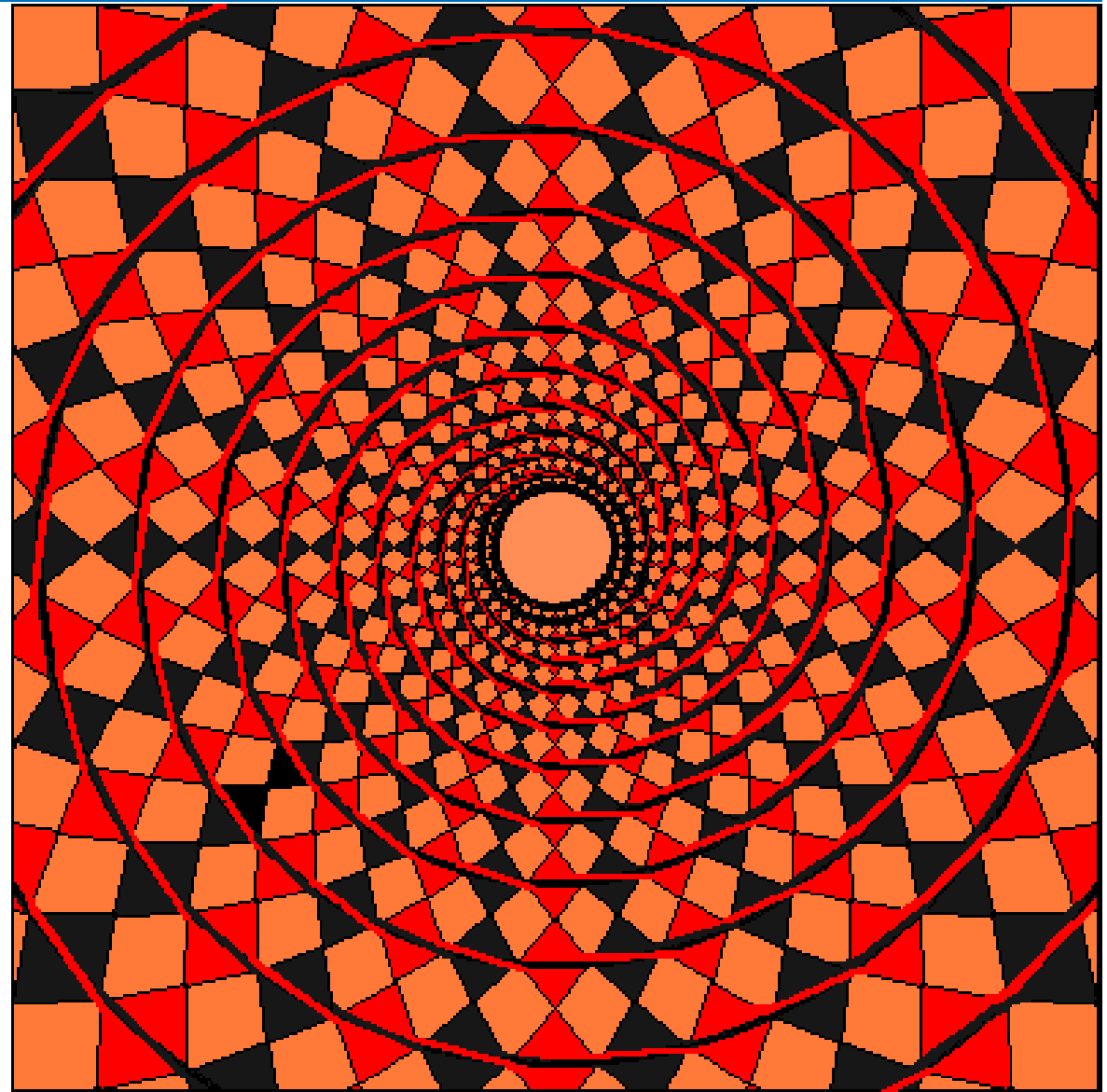
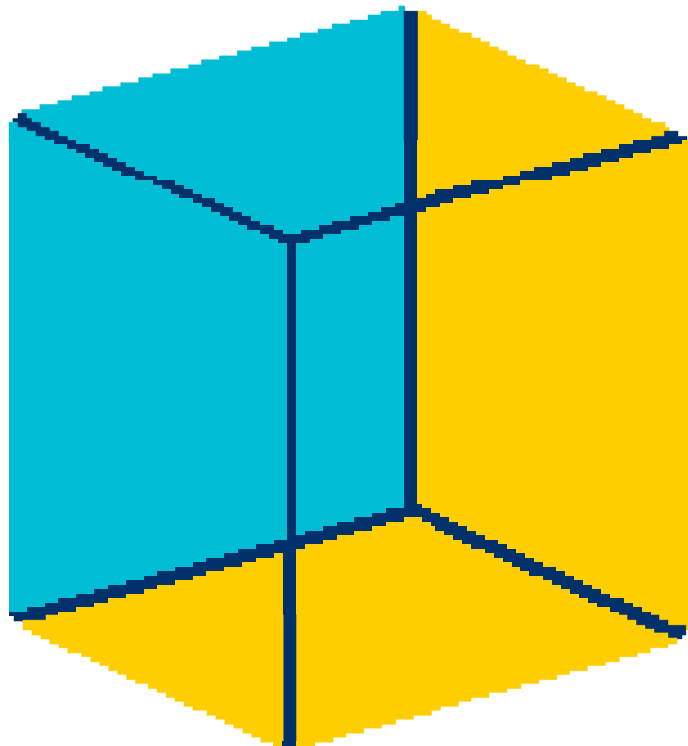


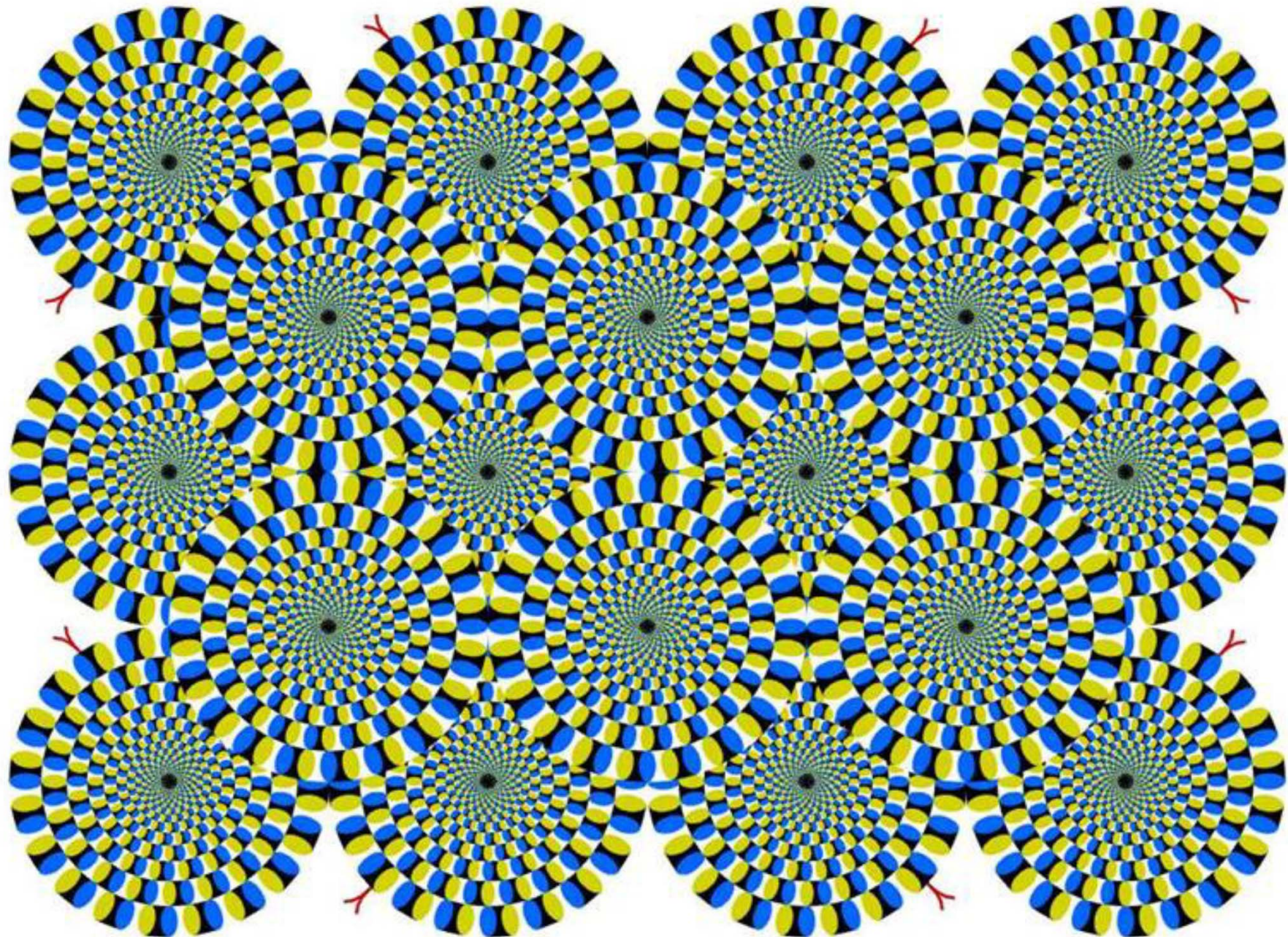
Optical Illusions (cont...)

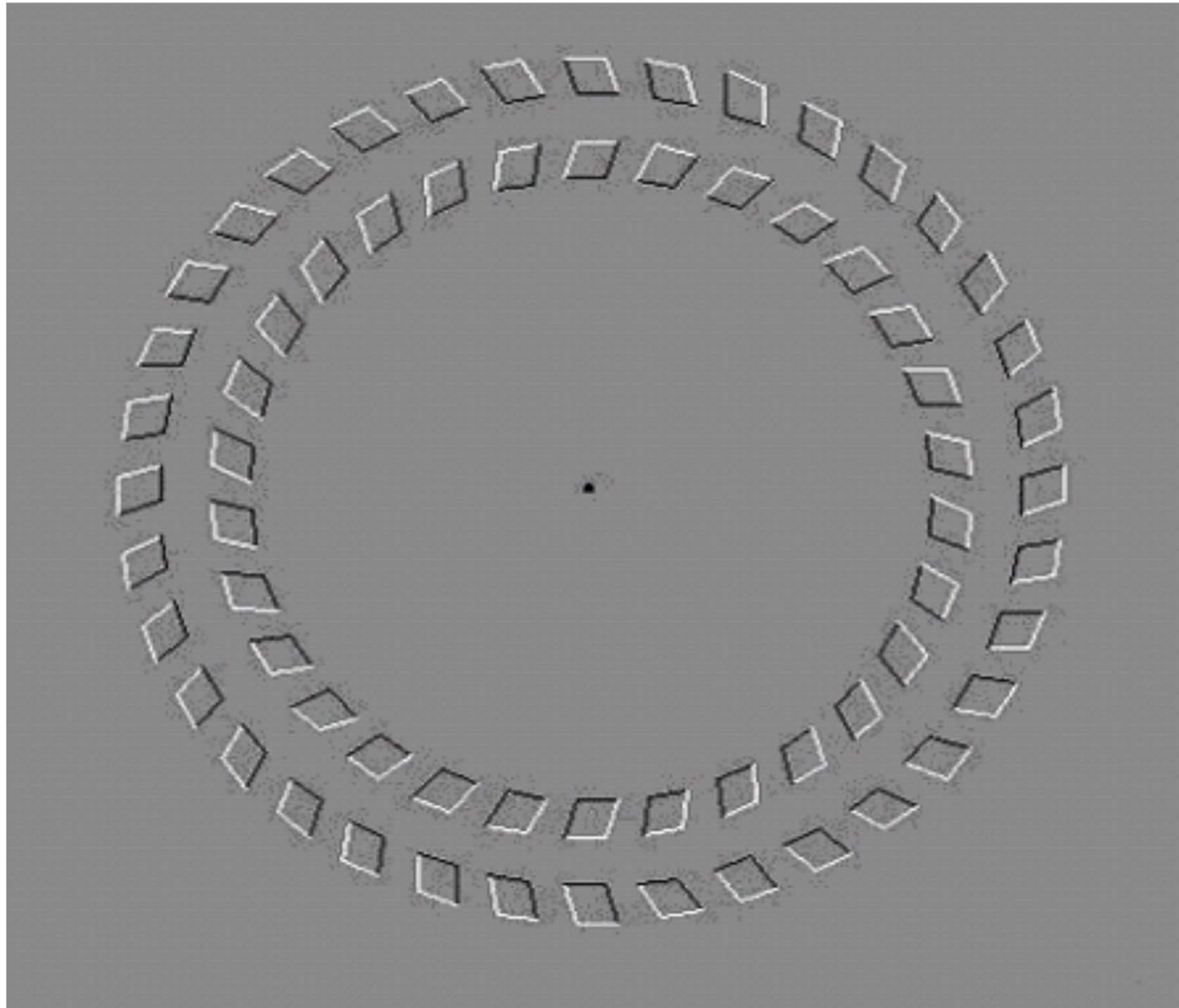


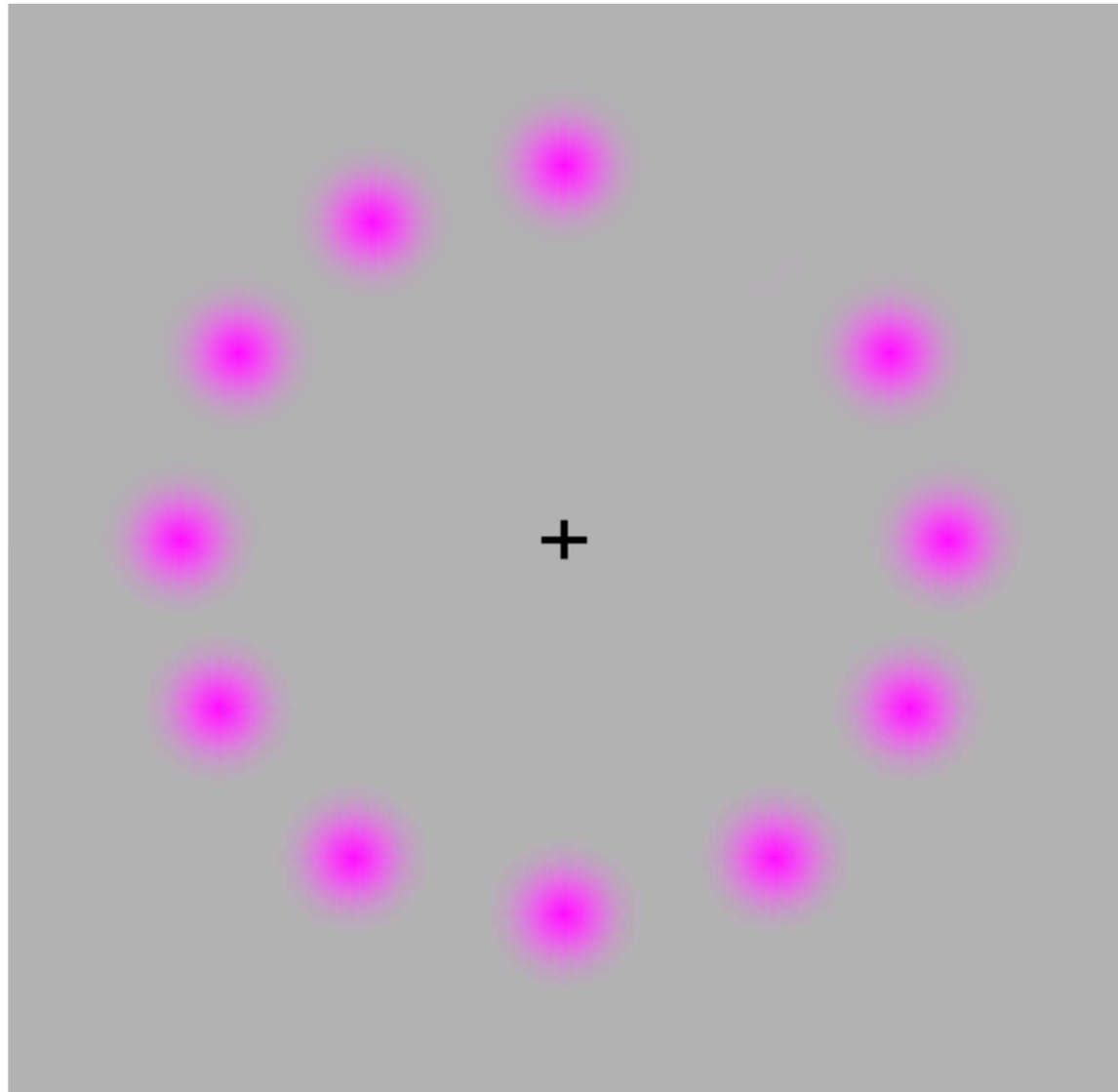
Stare at the cross
in the middle of
the image and
think circles





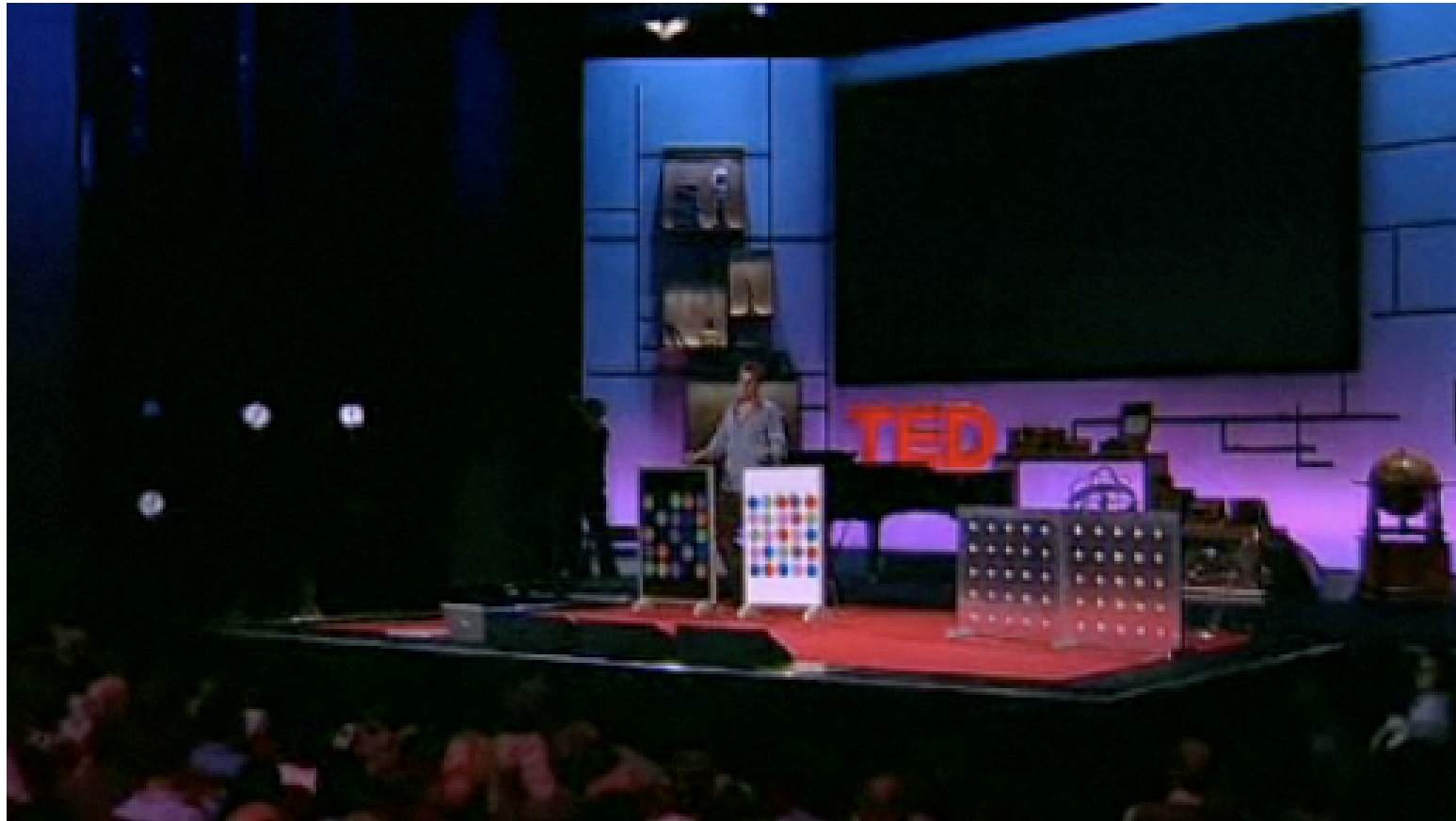






Concentrate on the cross in the middle, and the moving circle turns green! ... after a while the violet circles disappear!!

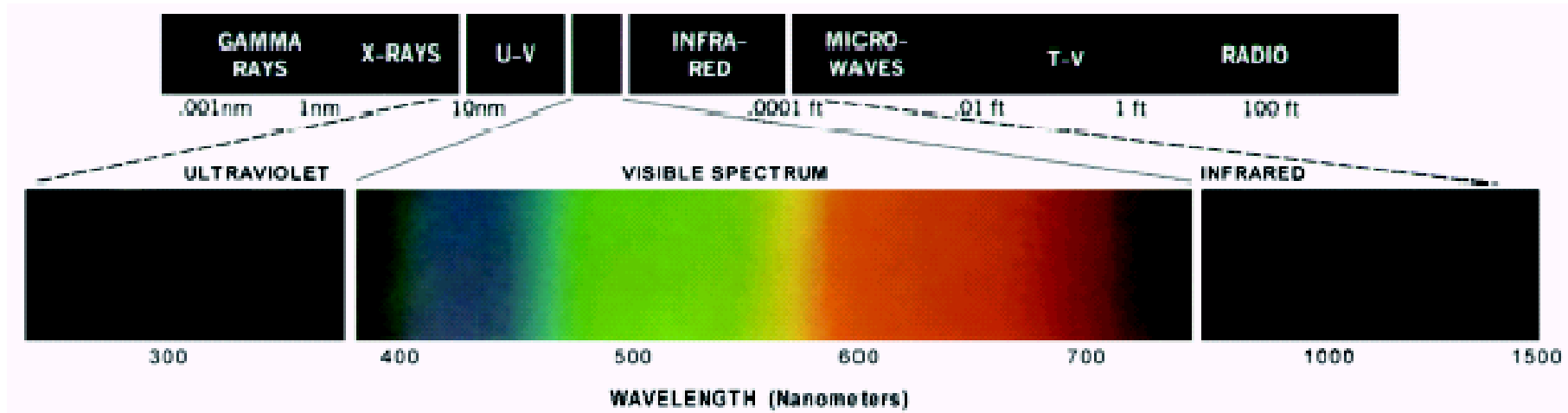
Mind Map Exercise: Mind Mapping For Note Taking



2 Light And The Electromagnetic Spectrum

Light is just a particular part of the electromagnetic spectrum that can be sensed by the human eye.

The electromagnetic spectrum is split up according to the wavelengths of different forms of energy.



Light: the Visible Spectrum

achromatic or monochromatic light

- We perceive only a small range of colors of the electromagnetic spectrum: $0.43\mu\text{m}$ (violet)- $0.78\mu\text{m}$ (red)
- Six bands: violet, blue, green, yellow, orange, red
- The color of an object is determined by the nature of the light *reflected* by the object
- Monochromatic light (gray level)
- Three elements measuring chromatic light
 - Radiance, luminance and brightness

Three basic quantities are used to describe the quality of a chromatic light source: radiance, luminance, and brightness

Radiance(unit: *W*) is the total amount of energy that flows from the light source

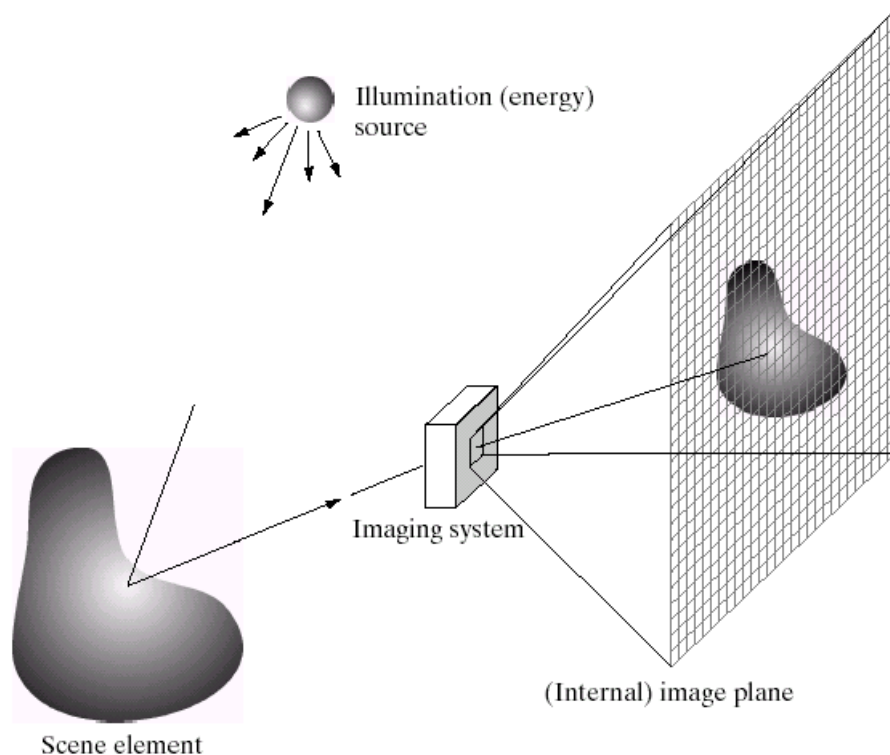
Luminance(unit: *lm*, *lumens*) gives a measure of the amount of energy an observer perceives from a light source

ex) Light emitted from a source operating in the far infrared region of the spectrum could have high radiance, but an observer would hardly perceive it(zero luminance)

Brightness is a subjective descriptor of light perception that is practically impossible to measure. It embodies the achromatic notion of intensity

3 Image sensing and acquisition

Images are typically generated *by illuminating a scene and absorbing* the energy reflected by the objects in that scene

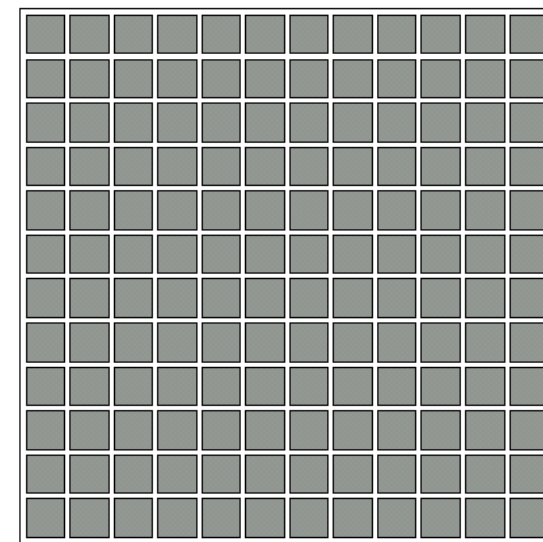
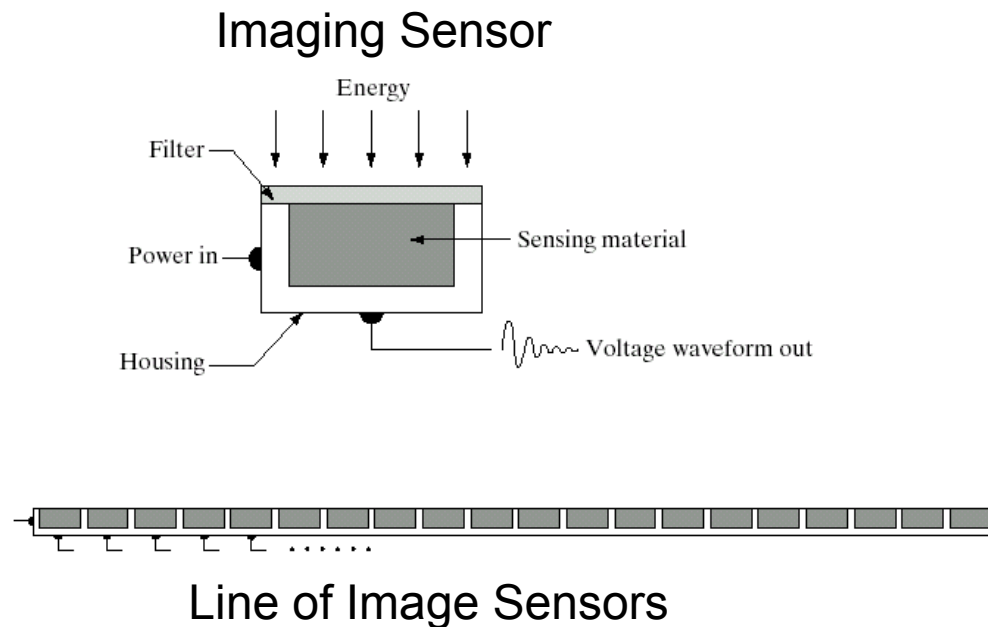


Typical notions of illumination and scene can be way off:

- a) X-rays of a skeleton
- b) Ultrasound of an unborn baby
- c) Electro-microscopic images of molecules

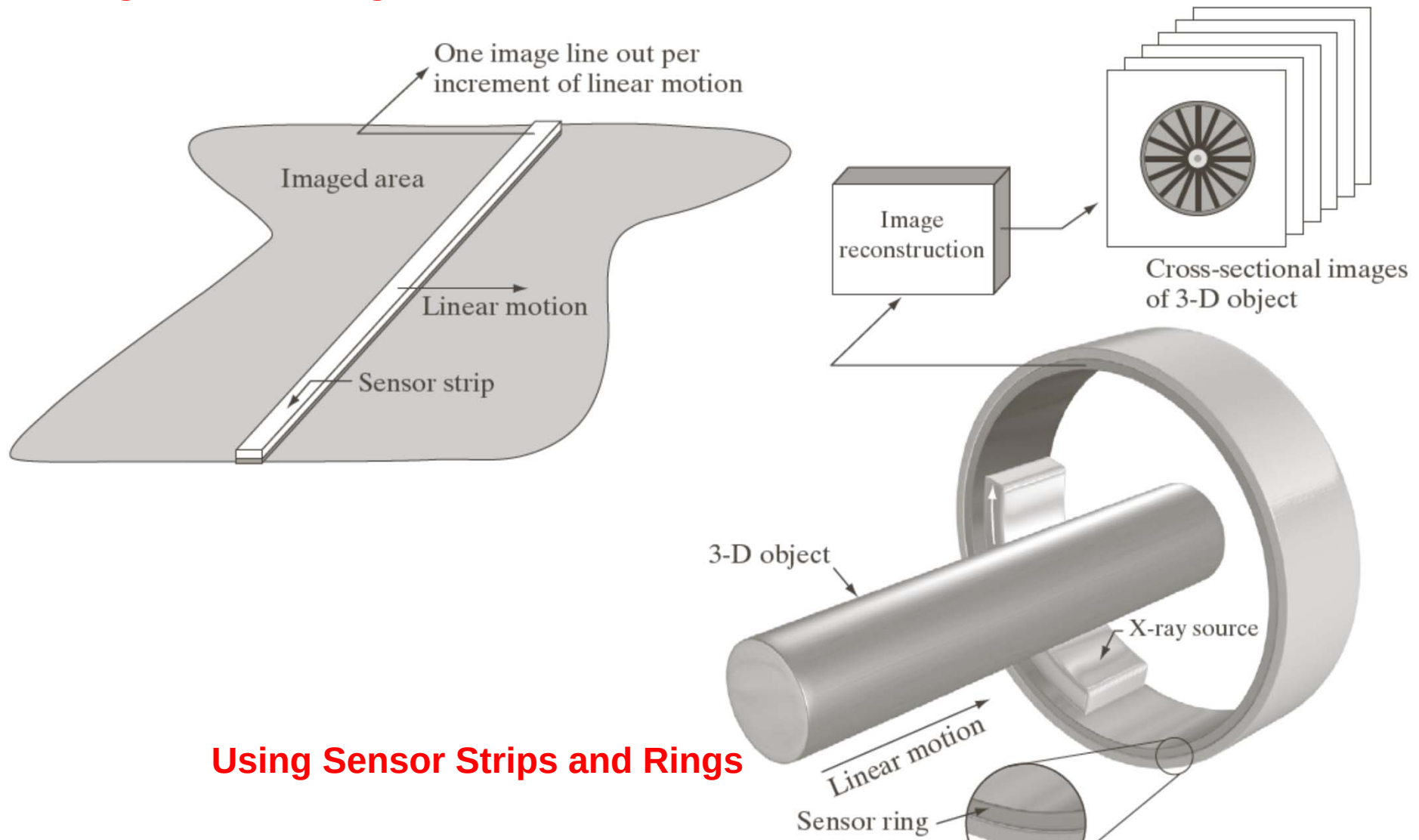
1) Image Sensing

Incoming energy is transformed into a voltage by the combination of input electrical power and sensor material that is responsive to the particular type of energy being detected. Three sensor arrangements:



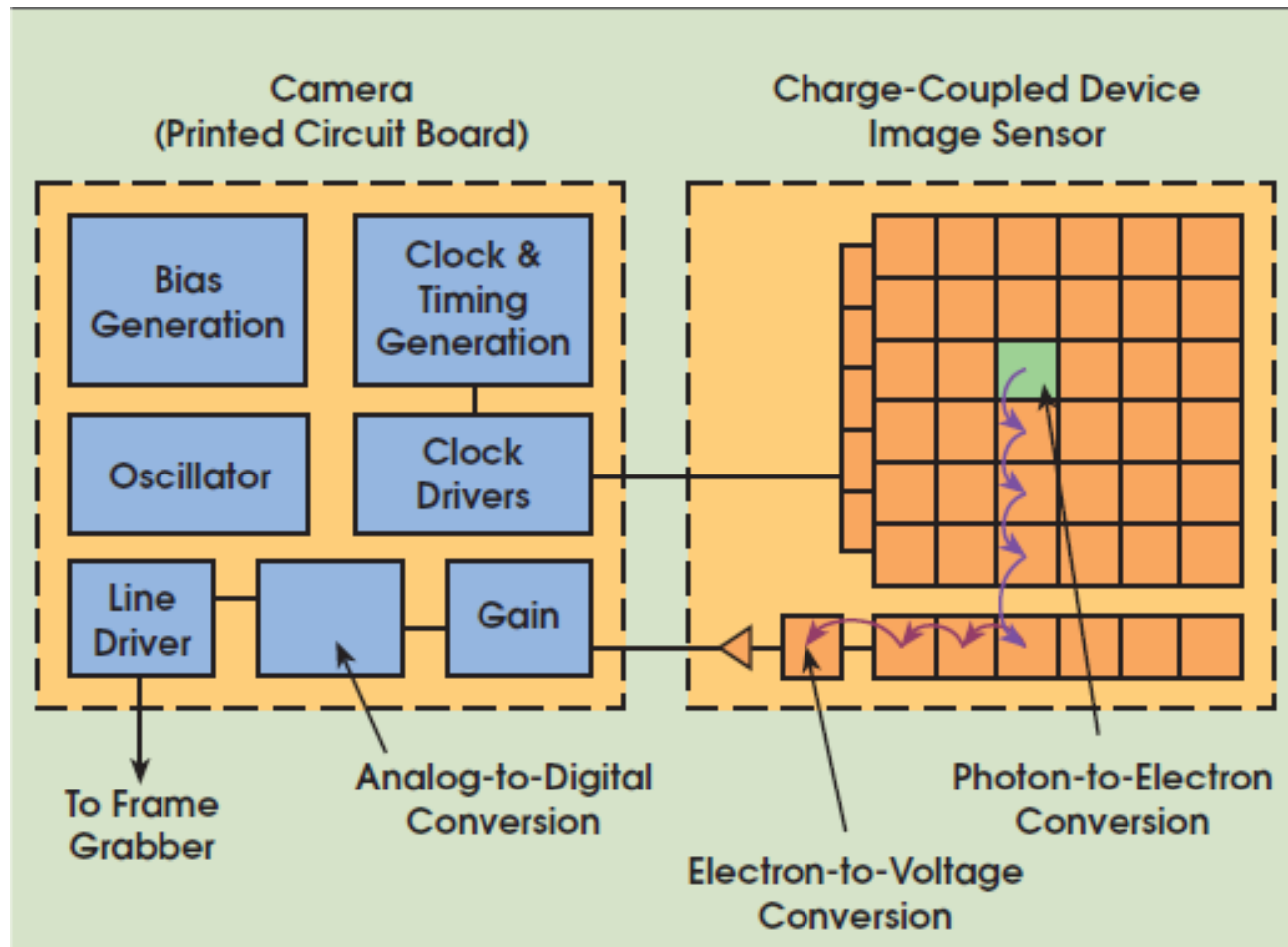
Array of Image Sensors

Image Sensing



Using Sensor Strips and Rings

Charge coupled device (CCD) image sensor





The Nobel Prize in Physics 2009

"for groundbreaking achievements concerning the transmission of light in fibers for optical communication"

"for the invention of an imaging semiconductor circuit – the CCD sensor"



Photo: U. Montan

Charles K. Kao



Photo: U. Montan

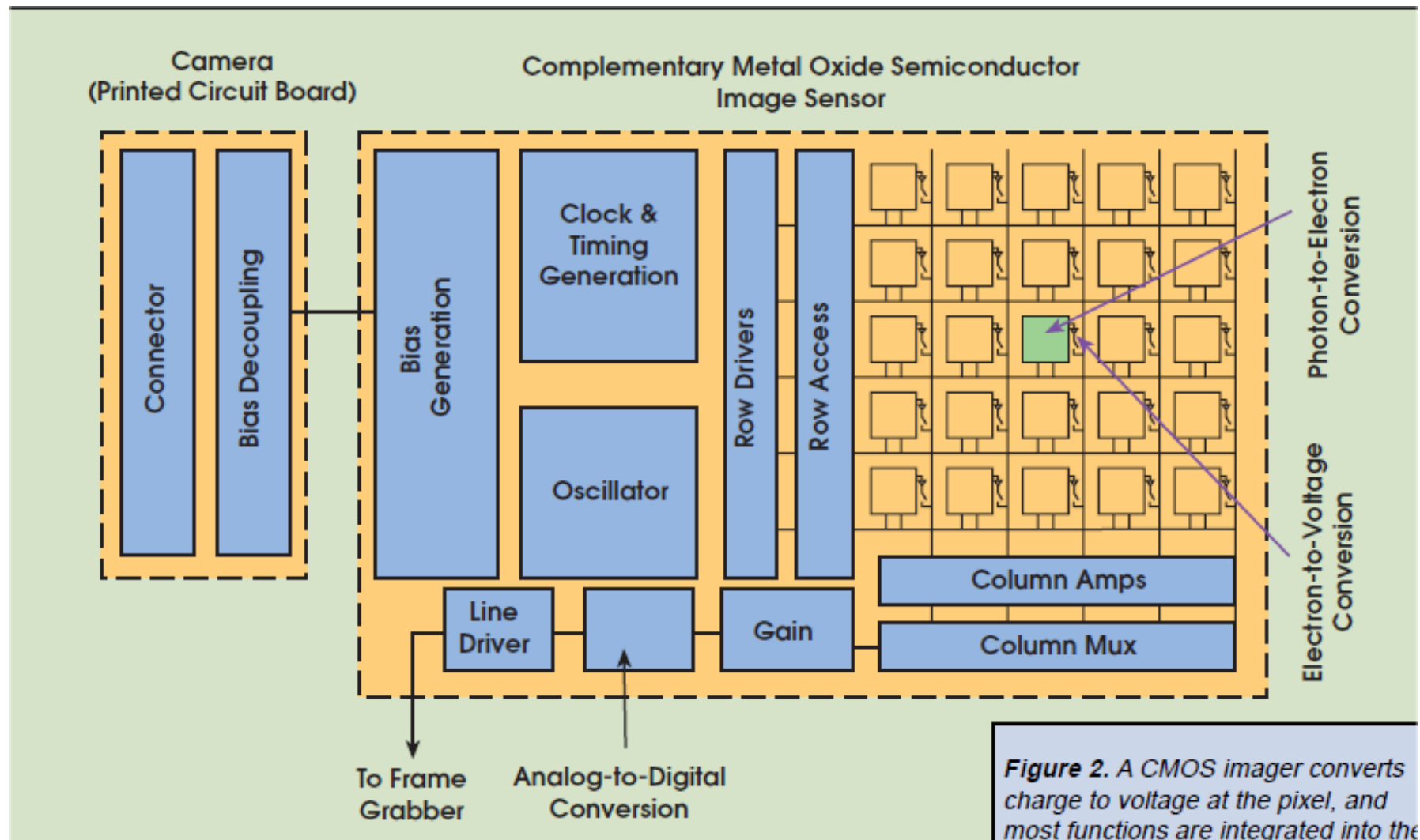
Willard S. Boyle



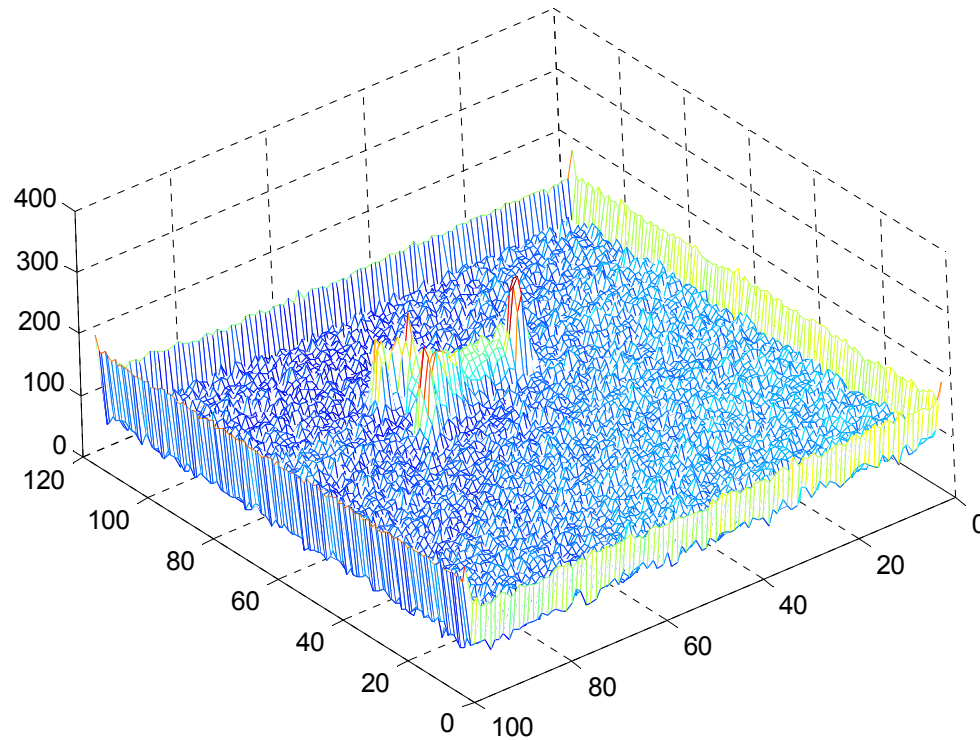
Photo: U. Montan

George E. Smith

Complementary Metal Oxide Semiconductor (CMOS) Sensor



3D Visualization



It is useful to take an analogy to rain gauge (image intensity values Measure the amount of ``photon rain’’)

Digital Single-Lens Reflection (DSLR) Cameras



High Dynamic Range Imaging



HDR Display (After Toner Mapping)



Thermal Imaging

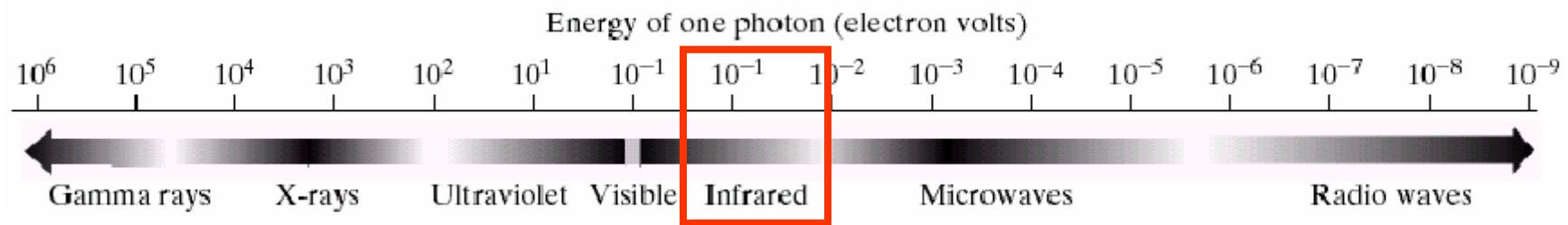
Operate in infrared frequency



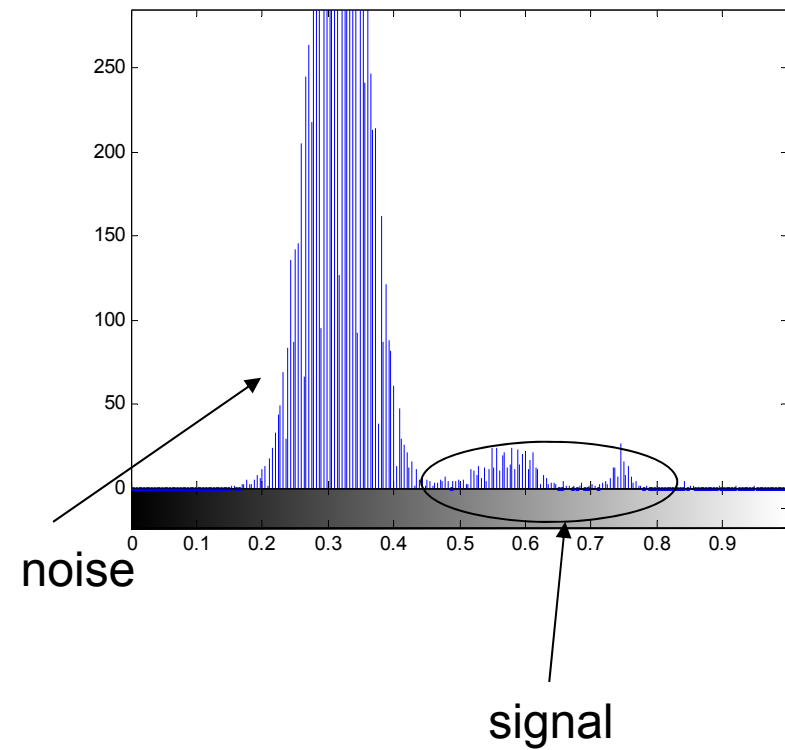
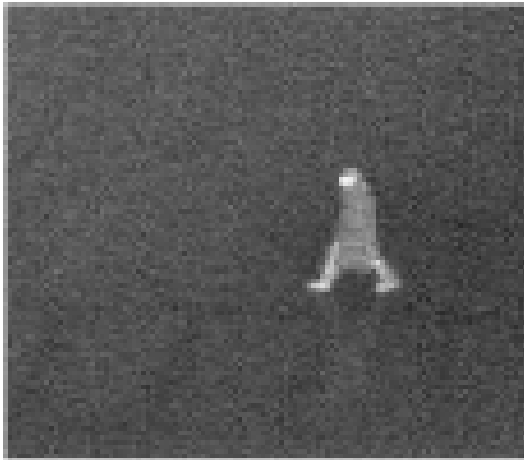
Grayscale representation
(bright pixels correlate with
high-temperature regions)



Pseudo-color representation
(Human body dispersing
heat denoted by red)

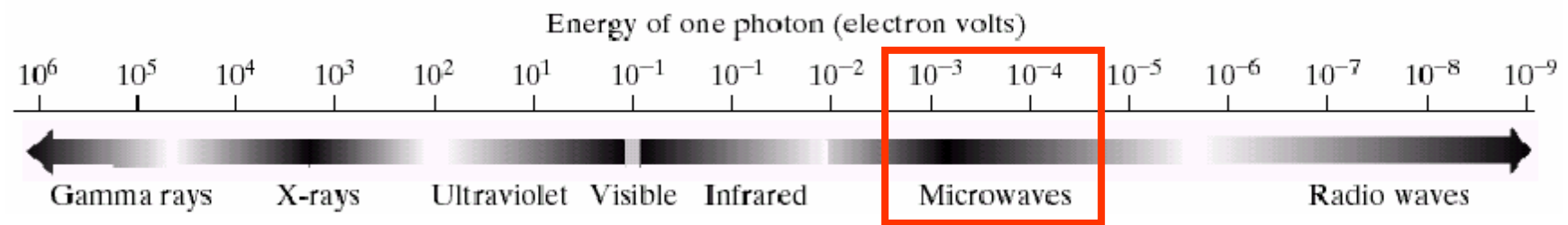
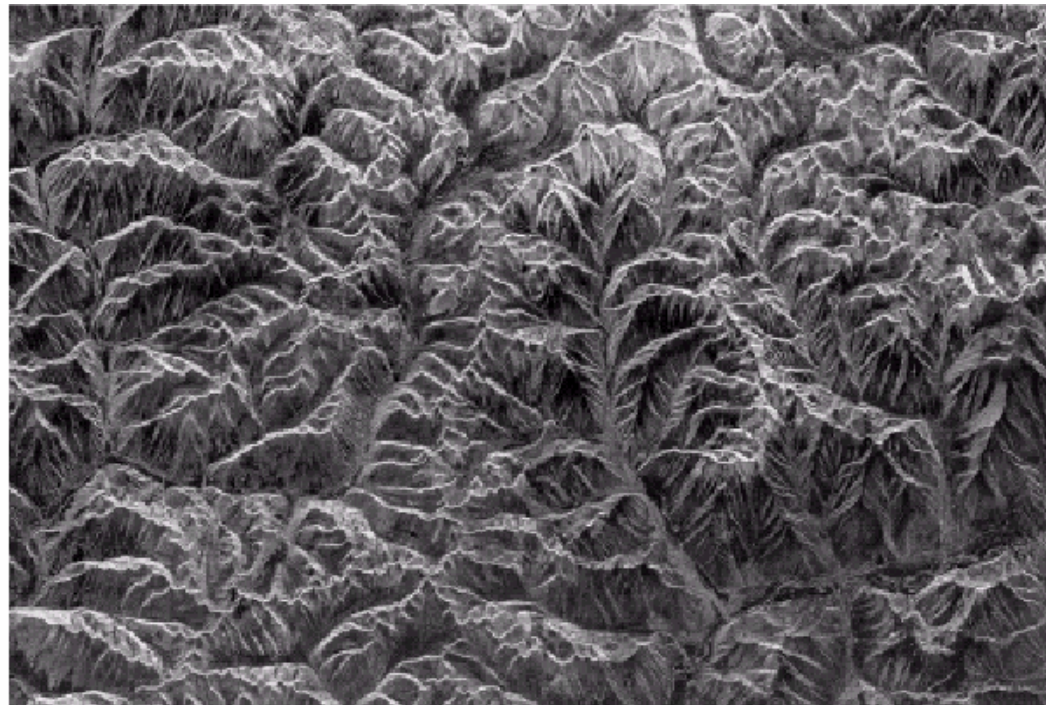


Low Signal-to-Noise (SNR) Behavior



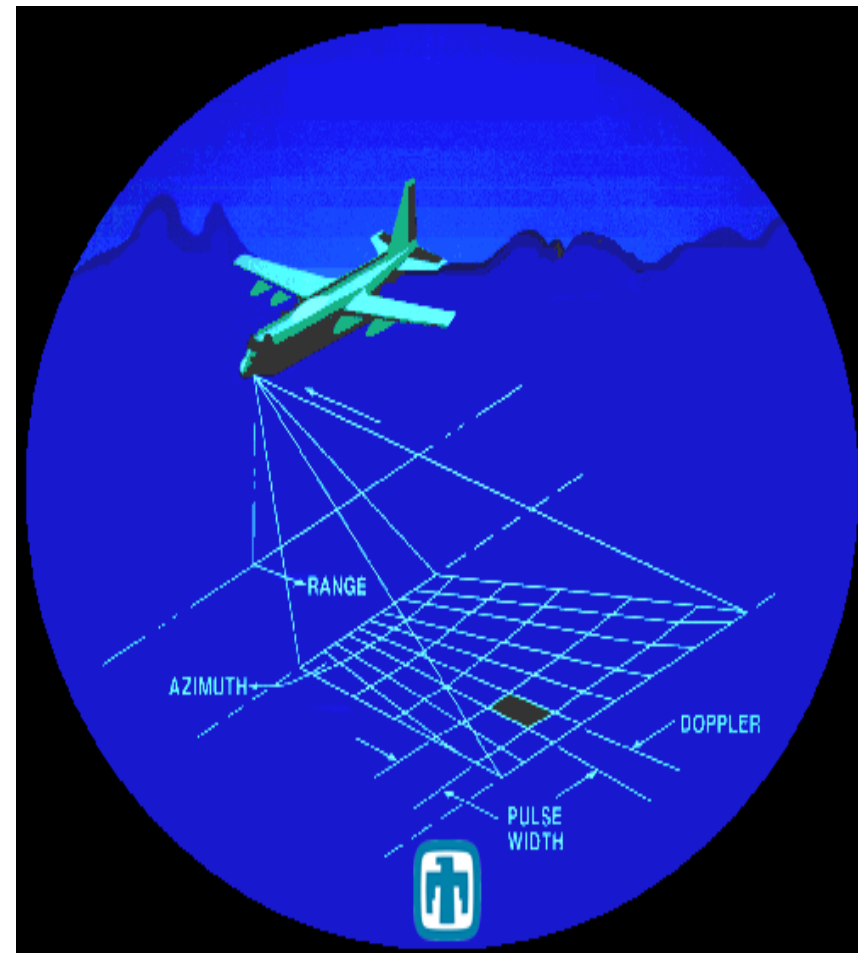
Radar Imaging

Operate in microwave frequency



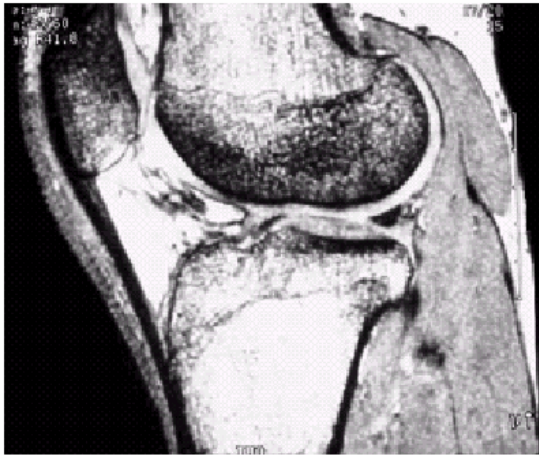
Synthetic Aperture Radar (SAR)

- Environmental monitoring, earth-resource mapping, and military systems
- SAR imagery must be acquired in inclement weather and all-day-all-night.
- SAR produces relatively fine resolution that differentiates it from other radars.



Magnetic Resonance Imaging (MRI)

Operate in radio frequency



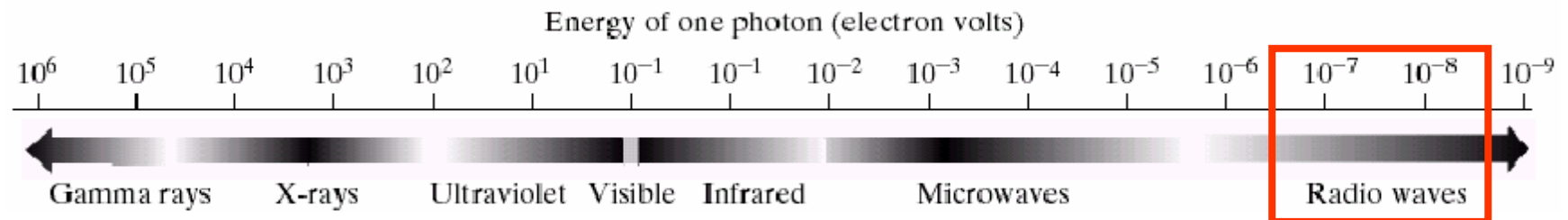
knee



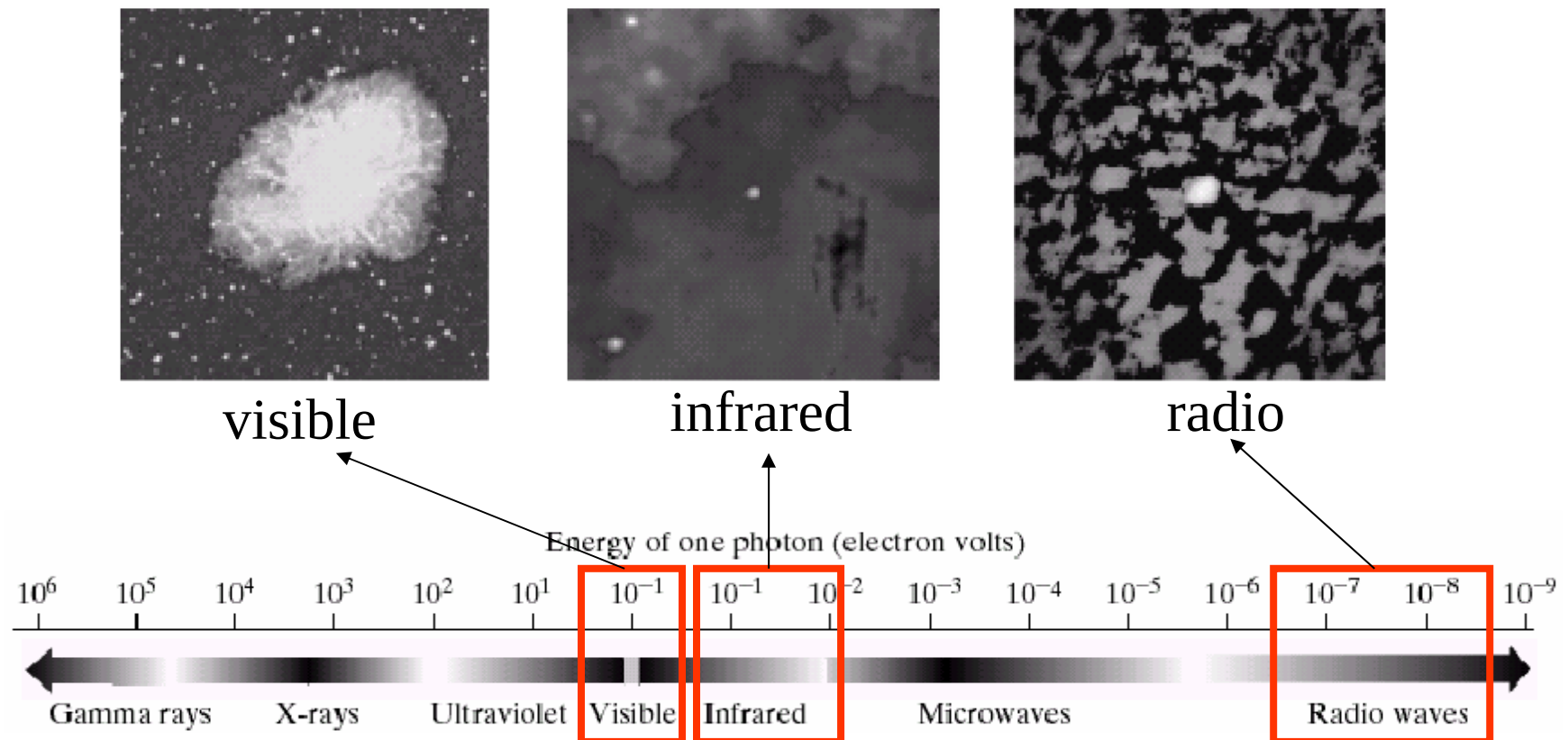
spine



head

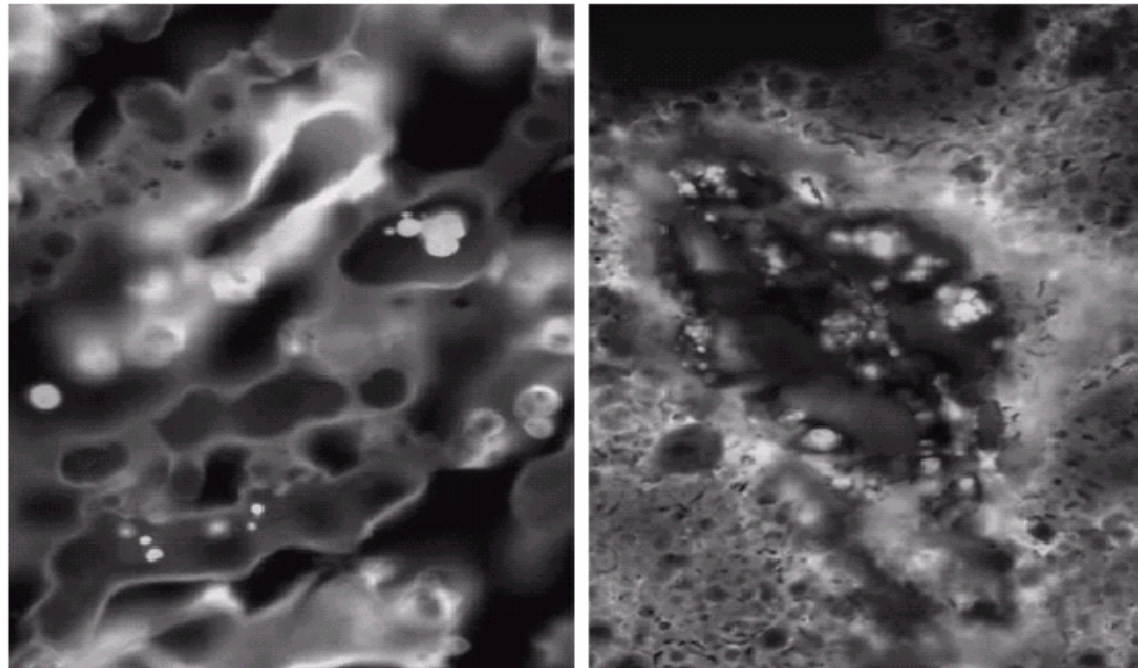


Comparison of Different Imaging Modalities



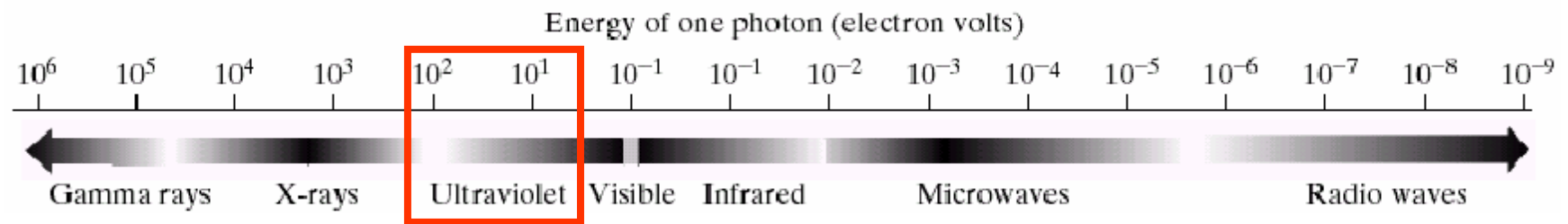
Fluorescence Microscopy Imaging

Operate in ultraviolet frequency

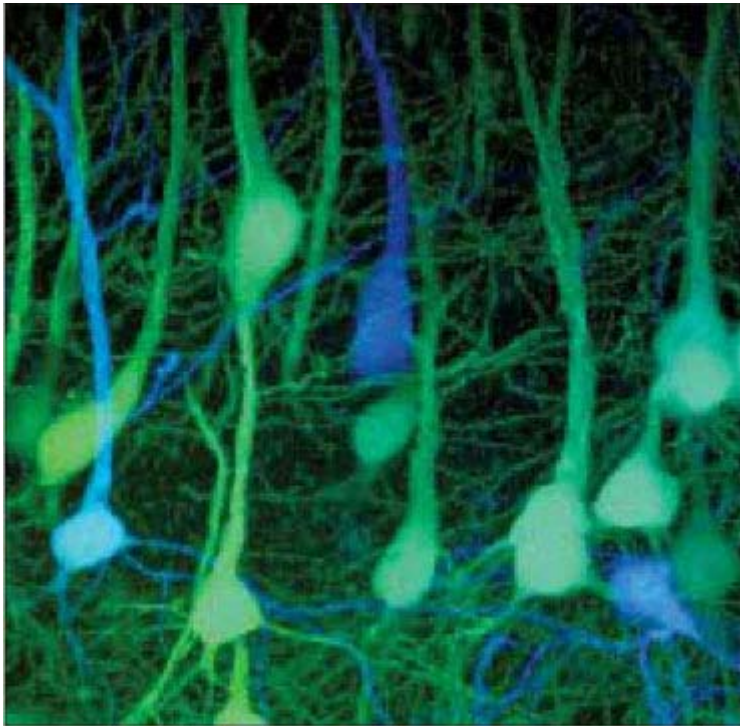


normal corn

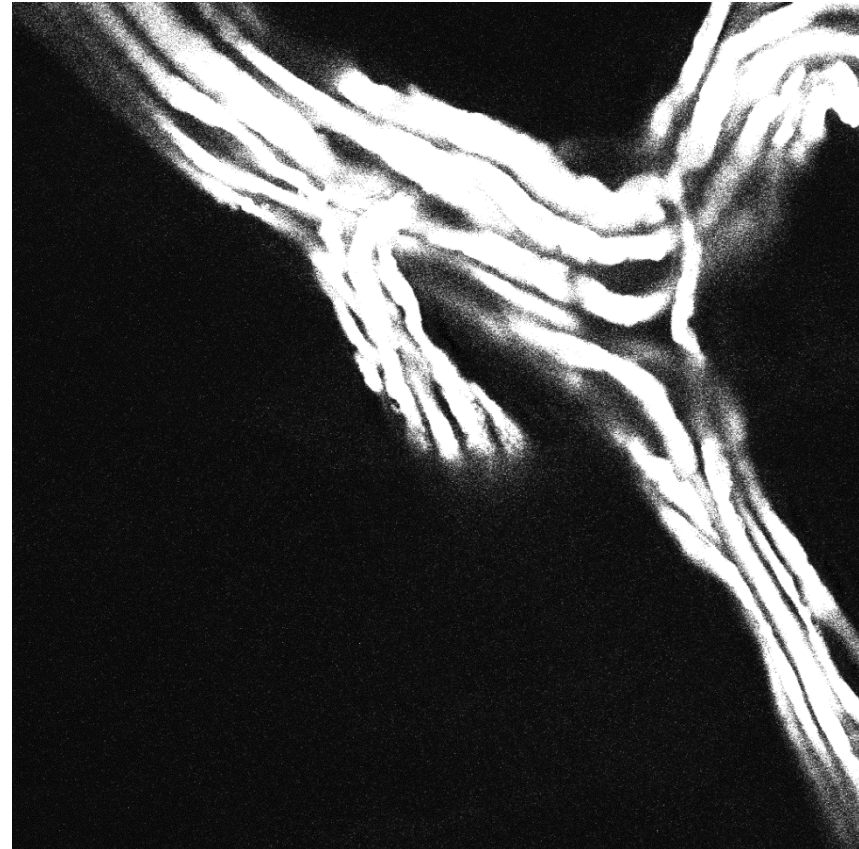
smut corn



What Does a Neuron Look Like?

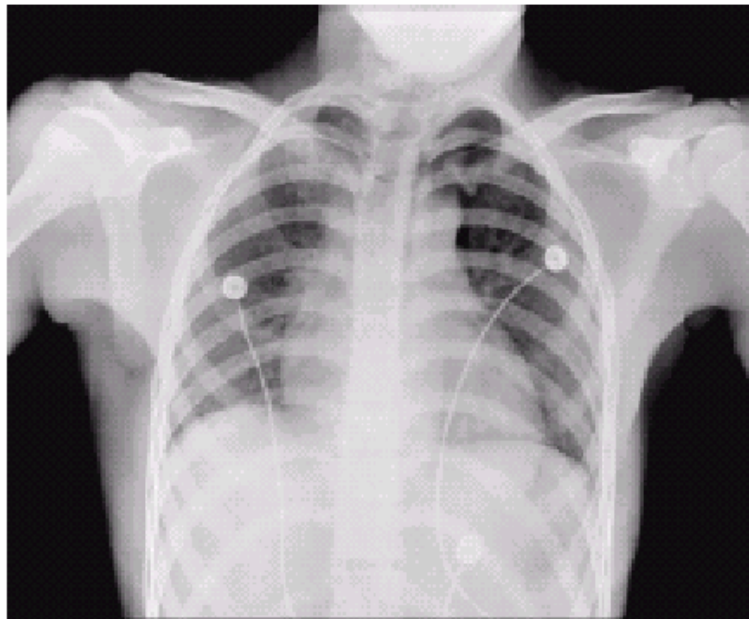


Artistic illustration

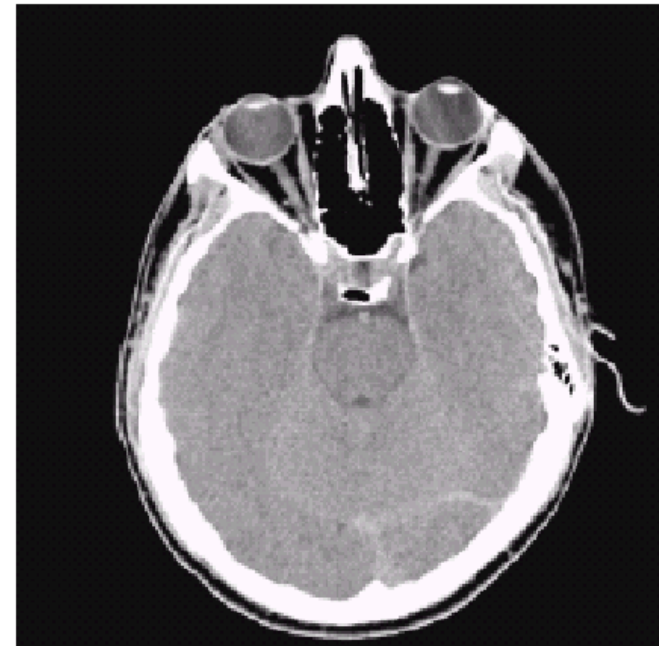


Real image

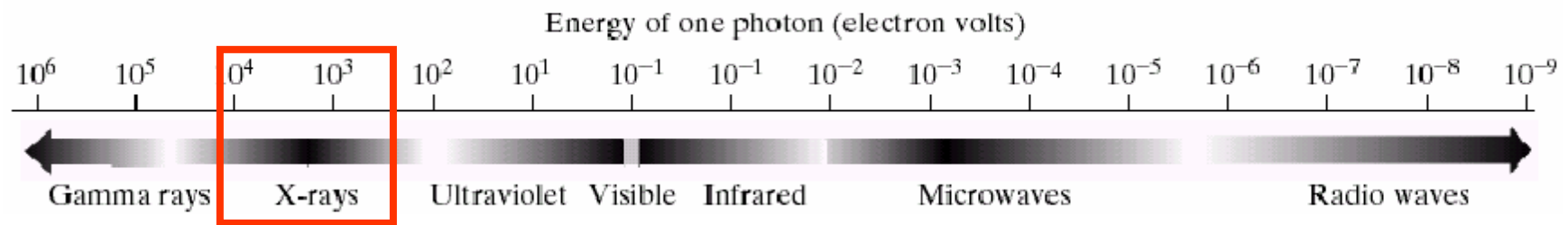
X-ray Imaging



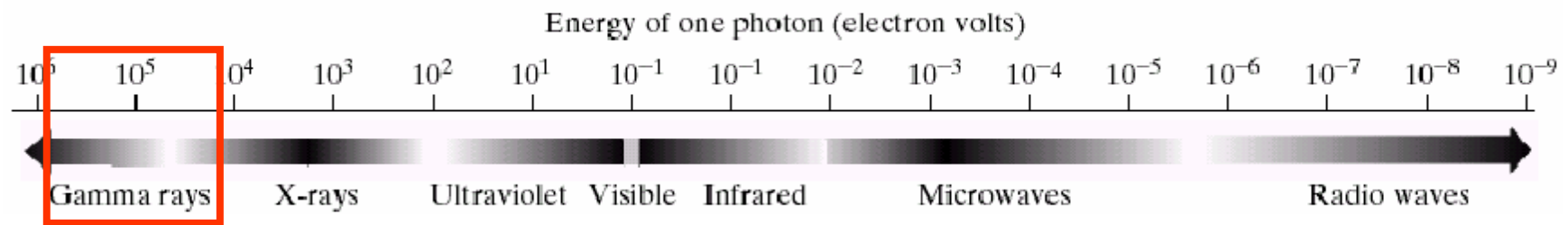
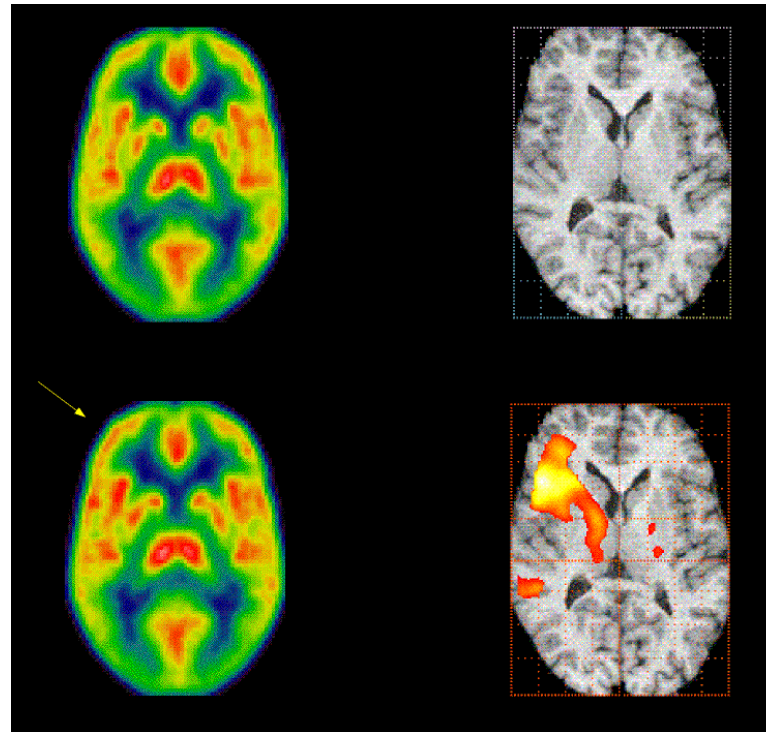
chest



head



Operate in gamma-ray frequency



2) Simple Image Formation Model

Monochrome(Gray) Image : $f(x,y)$

$f(x,y) =$ intensity value at coordinates (x,y)

$0 < f(x,y) < \infty$; $f(x,y)$ is energy

Simple image formation

$$f(x,y) = i(x,y)r(x,y)$$

$0 < i(x,y) < \infty$; illumination

$0 < r(x,y) < 1$; reflectance

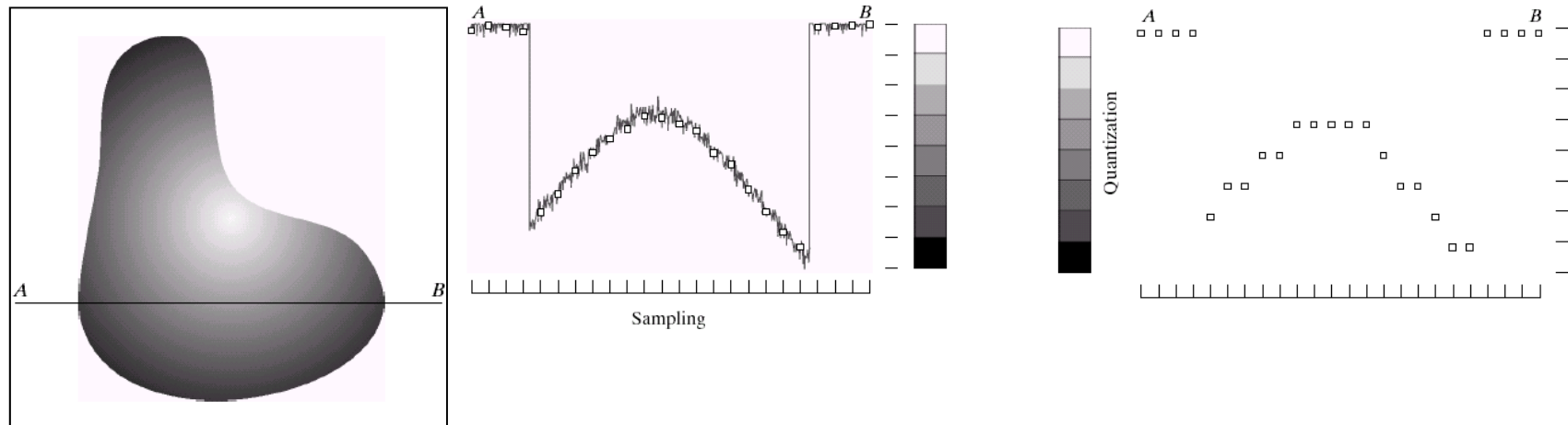
In real situation (level)

$$L_{\min} \leq I (=f(x,y)) \leq L_{\max}$$

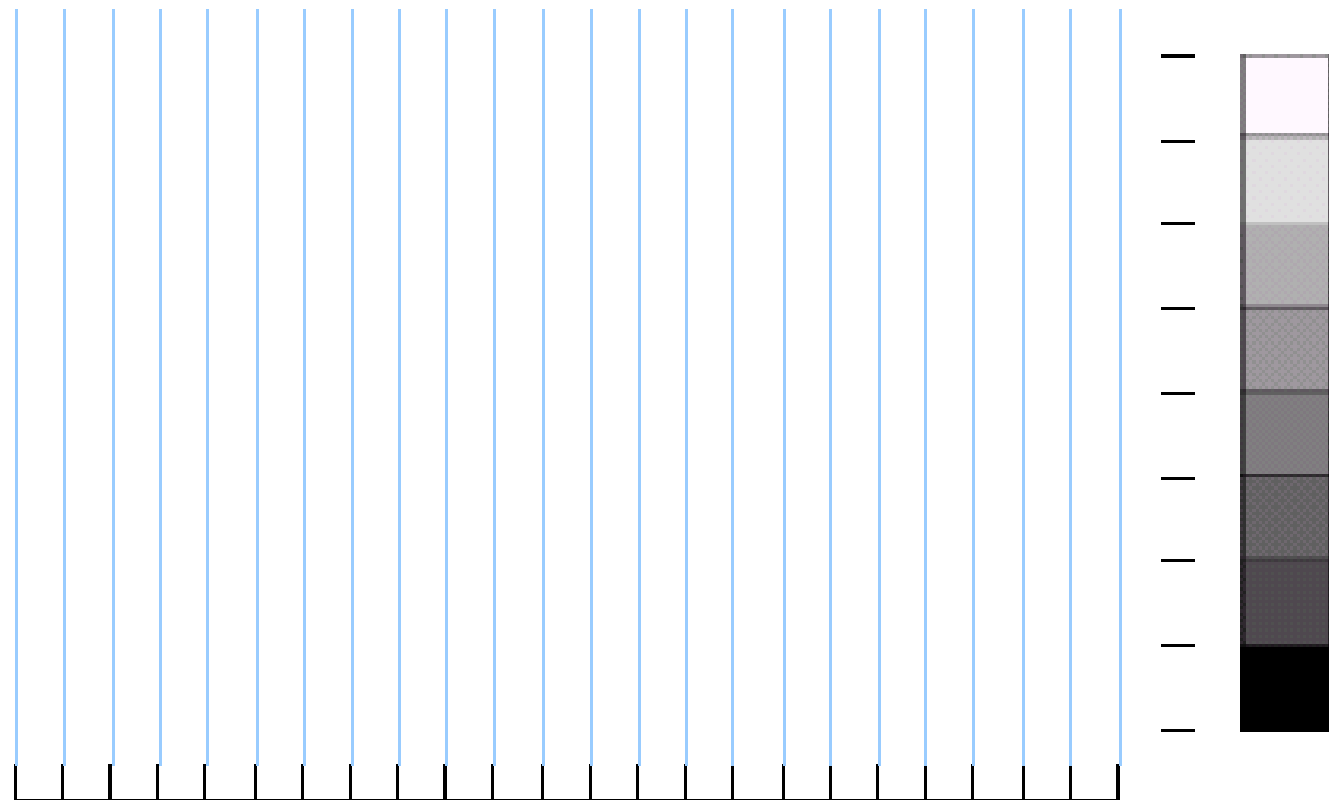
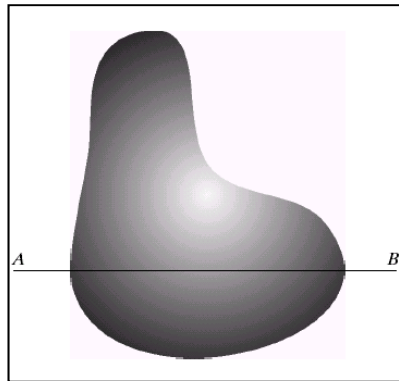
4 Image sampling, quantization and representation

A digital sensor can only measure a limited number of **samples** at a **discrete** set of energy levels

Quantisation is the process of converting a continuous **analogue** signal into a digital representation of this signal



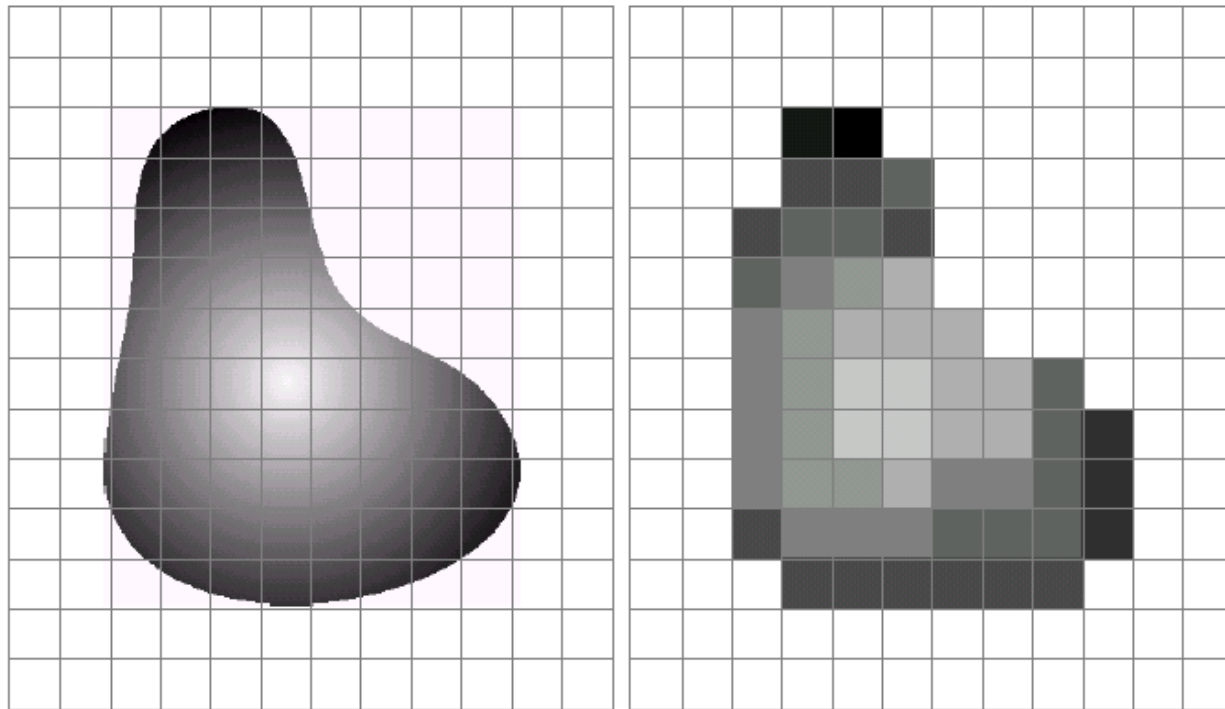
4.1 Image Sampling and Quantisation



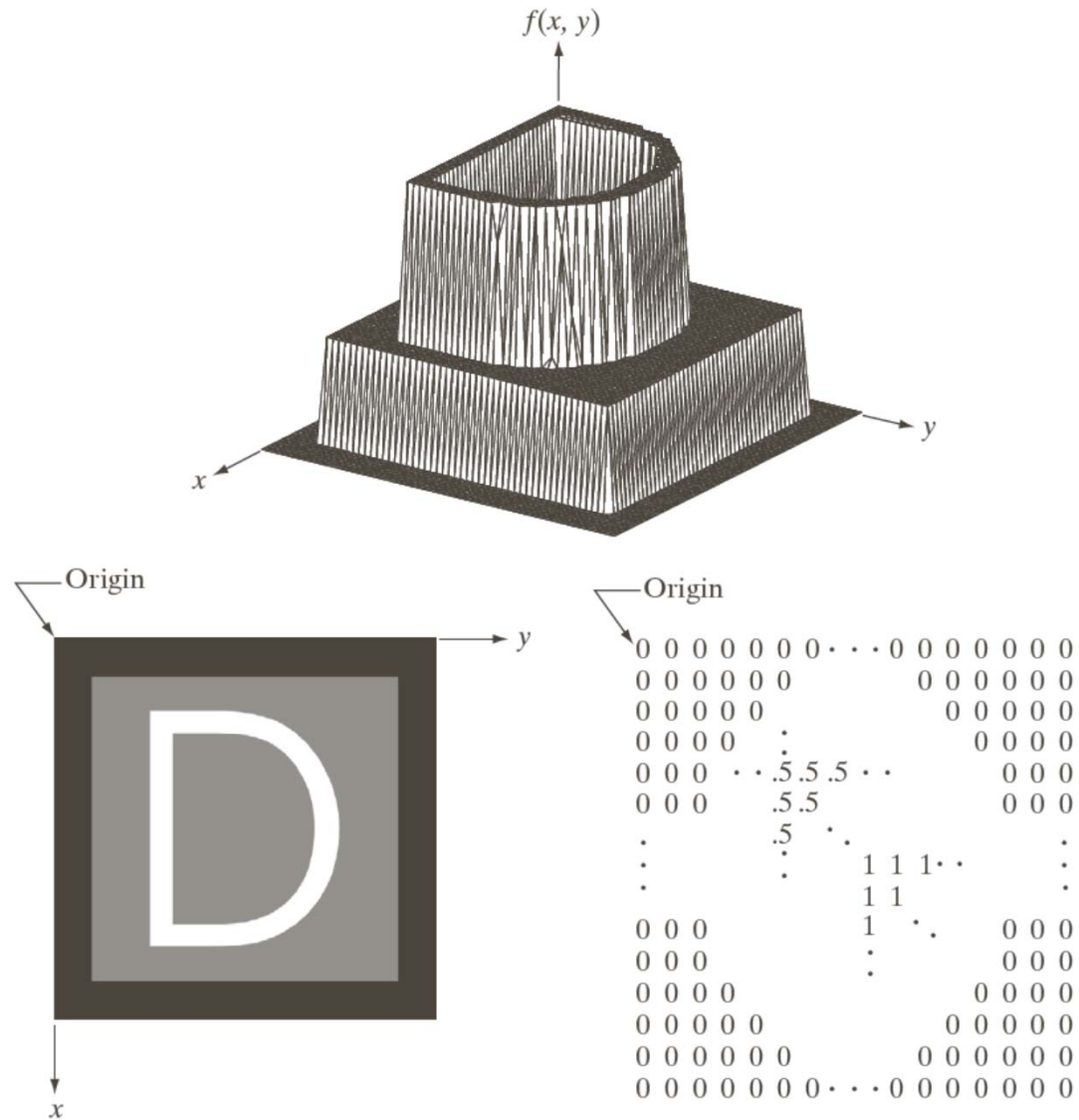
Sampling

4.1 Image Sampling and Quantisation

Remember that a digital image is always only an **approximation** of a real world scene



4.2 Image representation

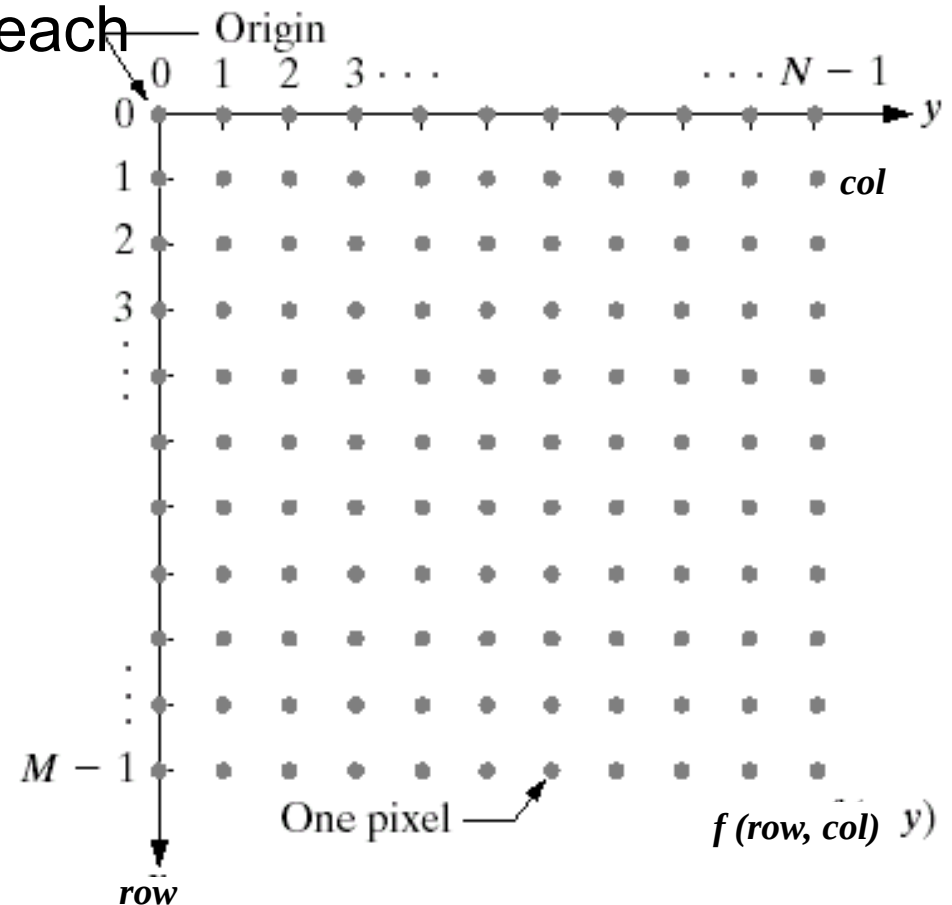


4.2 Image representation

A digital image is composed of M rows and N columns of pixels each storing a value.

Pixel values are most often grey levels in the range 0-255 (black-white).

We will see later on that images can easily be represented as matrices.



5 Image Resolution

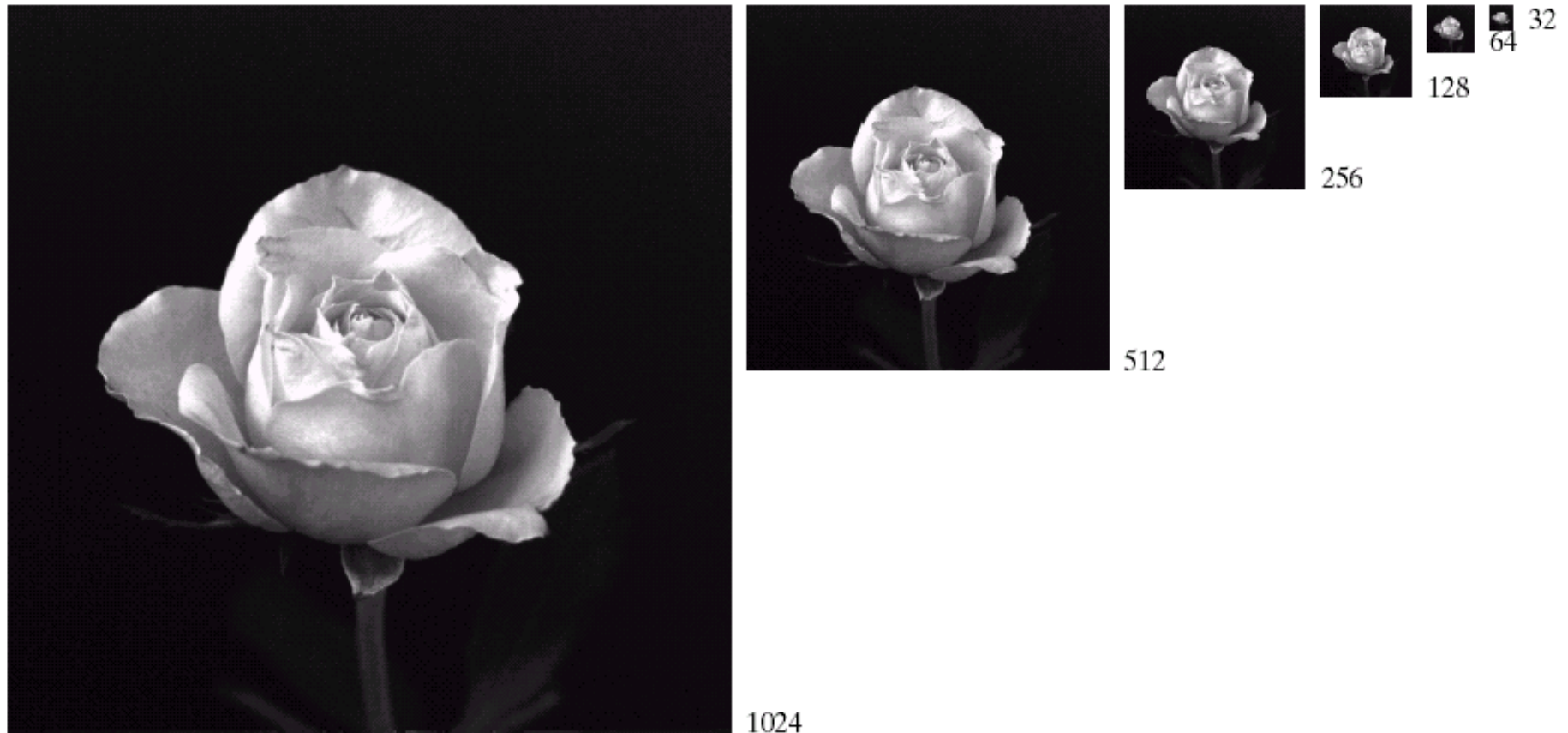
5.1 Spatial Resolution

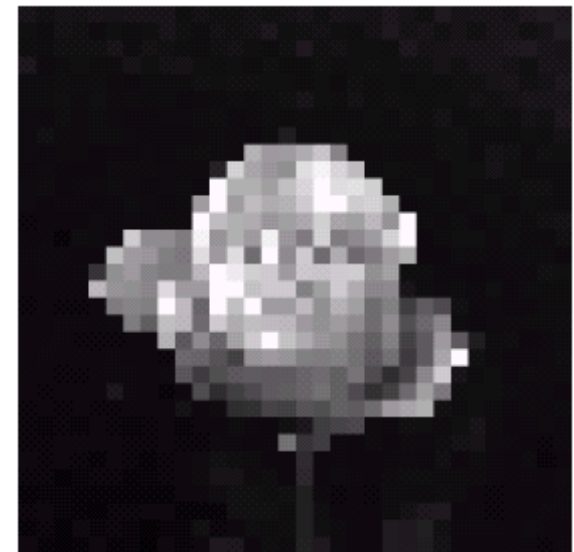
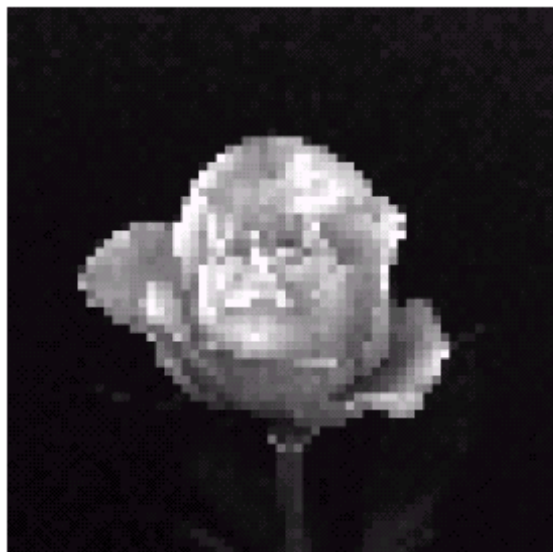
Spatial resolution of an image is determined by how sampling was carried out.

Spatial resolution simply refers to the smallest discernable detail in an image.

- Vision specialists often talk about pixel size
- Graphic designers will talk about *dots per inch* (DPI)

Spatial Resolution







original



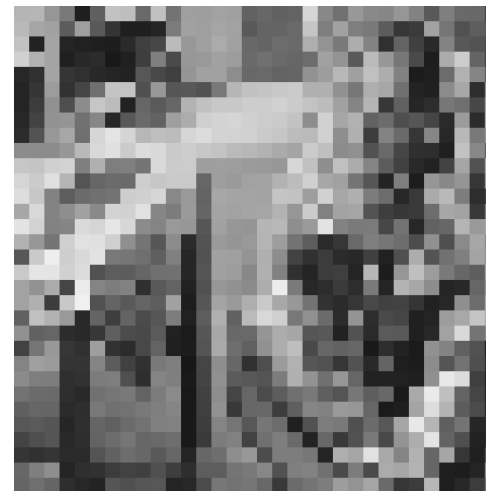
Sampled by 2



Sampled by 4



Sampled by 8



Sampled by 16

5.2 Intensity Level Resolution

Intensity level resolution refers to the number of intensity levels used to represent the image

- The **more** intensity levels used, the **finer** the level of detail discernable in an image
- Intensity level resolution is usually given in terms of the number of bits used to store each intensity level

Number of Bits	Number of Intensity Levels	Examples
1	2	0, 1
2	4	00, 01, 10, 11
4	16	0000, 0101, 1111
8	256	00110011, 01010101
16	65,536	1010101010101010

Intensity Level Resolution

256 grey levels (8 bits per pixel)



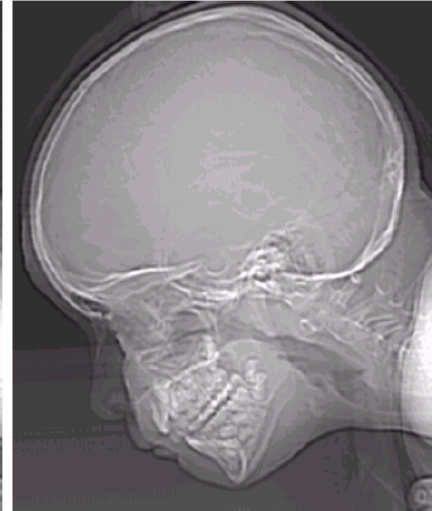
128 grey levels (7 bpp)



64 grey levels (6 bpp)



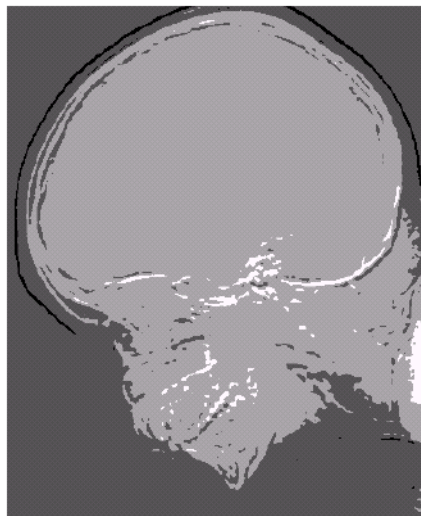
32 grey levels (5 bpp)



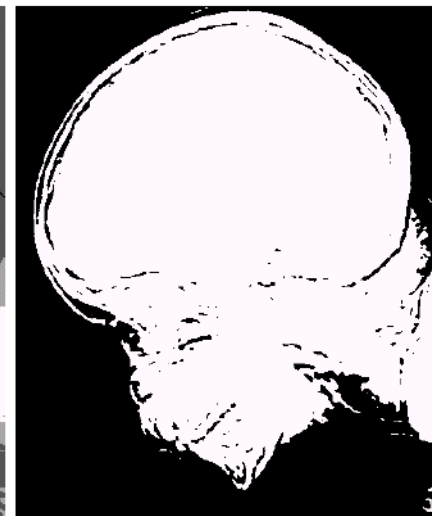
16 grey levels (4 bpp)



8 grey levels (3 bpp)



4 grey levels (2 bpp)



2 grey levels (1 bpp)



original



128 levels (7 bits)



16 levels (4 bits)



4 levels (2 bits)



2 levels (1 bit)

Resolution: How Much Is Enough?

- This all depends on what is in the image and what you would like to do with it
- Key questions include
 - Does the image look pleasing?
 - Can you see what you need to see within the image?

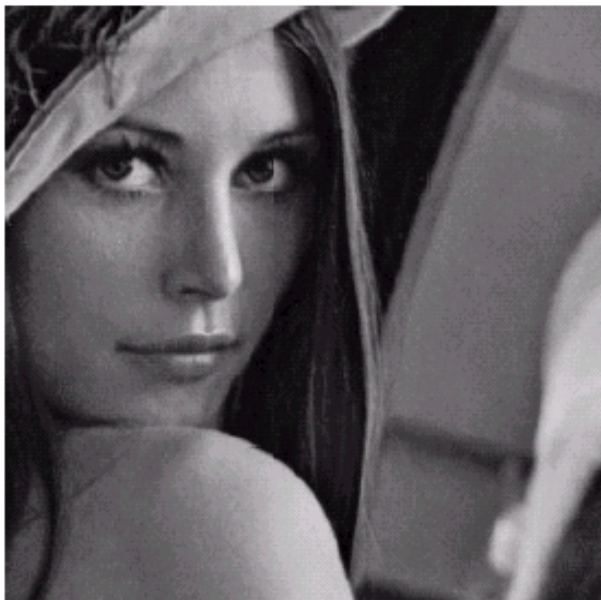


counting the number of cars?

reading the number plate?

Intensity Level Resolution

Images with a large amount of detail only a few intensity levels may be needed.



Low Detail



Medium Detail



High Detail

Summary

We have looked at:

- Human visual system
- Light and the electromagnetic spectrum
- Image sensing and acquisition
- Image representation
- Sampling, quantisation and resolution

Next time we start to look at techniques for image enhancement